



POLITECNICO
MILANO 1863

P.H.O.E.N.I. X

Programmable Hybrid Oceanic
Entity: Networked Inertial explorer

Bioinspired Robotics Lab

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A.Y. 2025-26

Professor Giovanni Bianchi¹

OUTLINE

01.

introduction



02.

Bioinspiration



03.

Kinematics
& dynamics



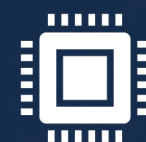
04.

Robotic
Fish
Design



05.

Electronics
& control



06.

Testing
& Results



01.

introduction

Motivation of the



Goal

Design and fabricate a prototype of a waterproof, buoyant robotic fish capable of sustained straight-line swimming



Bioinspired Design

- Replicate high-speed and efficient propulsion of **Atlantic Mackerel fish**
- **Carangiform** (Body-Caudal Fin) locomotion



Testing & Validation

- **Submerge** in a controlled aquatic environment
- Experimentally test the robot's **swimming speed and thrust generation**



Results and Discussion

- **Compare** experimental **results** with theoretical **models**
- Identify hydrodynamic and structural **design limitations**

o1 Why a Robotic Fish?

A bioinspired robotic fish supports marine research and inspection in a non-intrusive manner

Bioinspired Advantages

- Offers silent, maneuverable propulsion
 - **Minimal acoustic and wake signatures** compared to traditional propellers
 - Ideal for **non-disruptive** operations

Real-World Applications

- **Navigate and inspect** environments and infrastructure
 - **Monitor** coral reefs, water quality, pipelines
 - Support marine **research** and aquaculture operations without causing environmental stress

o1 Design Constraints &

Several design constraints and requirements must be adhered to during the design of the fish

Requirements



Manufacturing Limitations

- Fabricated exclusively using accessible **rapid-prototyping materials**
 - 3D-printed PLA/ABS for rigid parts
 - Silicone molding for flexible seals



Electronics & Actuation

- Logic and control driven by standard **Arduino** microcontrollers
- **Motors** specifically **sized based on** dynamic thrust requirements



Buoyancy & Stability

- Must achieve near-neutral **buoyancy**
- Prevent tilting/rolling using passive internal **counter-weights** and **secondary stabilizing fins**



Environmental Sealing

- **Complete waterproofing** of the internal electronics compartment
 - Critical to prevent failure during submerged operation

02.

bioinspiratio

o2 Biological Inspiration of

The **Atlantic Mackerel** was selected as inspiration for the robot and adapted to a practical prototype



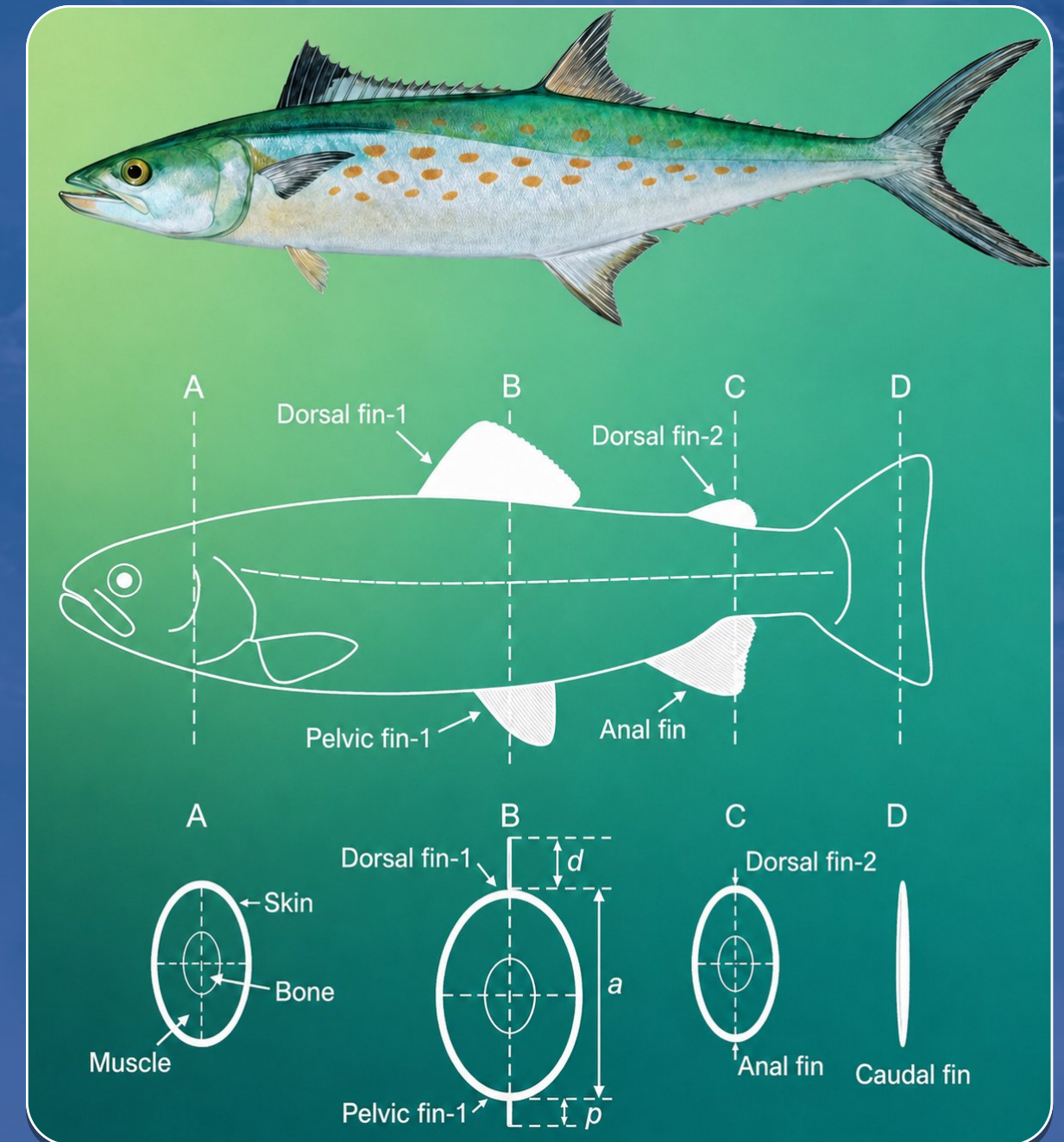
**Selected Species:
Mackerel Tuna**

- Inspired by the **Atlantic Mackerel**
 - Chosen for its high-speed cruising and exceptional propulsive efficiency
- Primary **inspiration** for the shape and dynamics of the prototype robot



**Geometric Scale &
Proportions**

- Total prototype model length scaled to **approximately 45.2 cm**
 - Sized to yield **adequate internal space** for the electronics and drive mechanism
 - Preserved **biological fin-to-body ratios** to match natural fluid dynamics

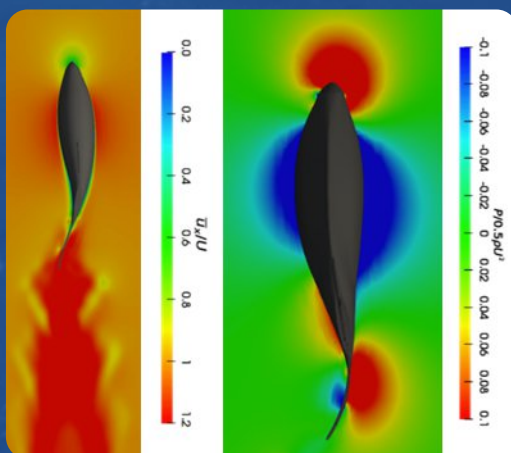


o2 Locomotion Strategy

The fish utilizes a **BCF Carangiform** locomotion strategy for thrust generation, steering, and stability

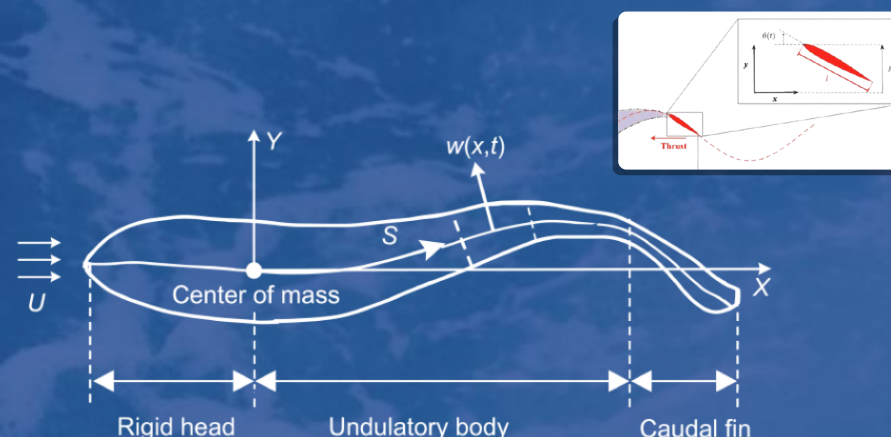
Body-Caudal Fin (BCF) Carangiform Swimming

Thrust Generation



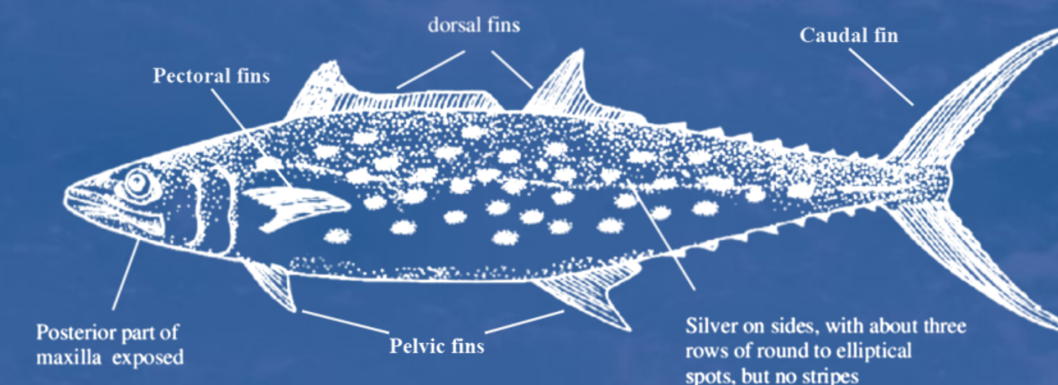
- **Synchronized action** of posterior body undulations and tail movements
- Combines side-to-side **sweeping** of fin (heaving) with angular **twisting** (pitching)
- Pushes water backward to generate a positive **net forward force** and overcome hydrodynamic drag

Undulatory Wave Profile



- **Bending** motion is strictly **localized** to the **posterior** section of the fish
 - Kinematic wave actions are confined to the **rear third** of the total body length
- This tail structure and movement strategy generates a **smooth traveling wave**

Hydrodynamic Fins

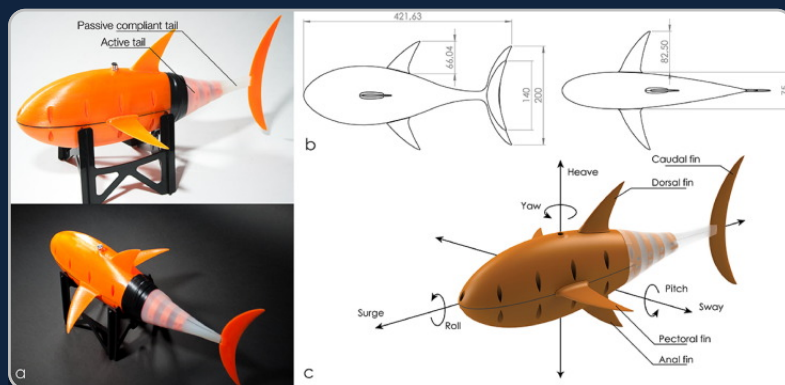


- Stiff, lunate-shaped **caudal fin** designed for optimized thrust transmission
- Passive secondary fins handle **orientation** (pectoral) and lateral **stability** control (dorsal/pelvic)

o2 From Biology to

To segment and adapt the fish to a robot, several **simplifications** were made to the morphology of the fish

Reference Inspiration



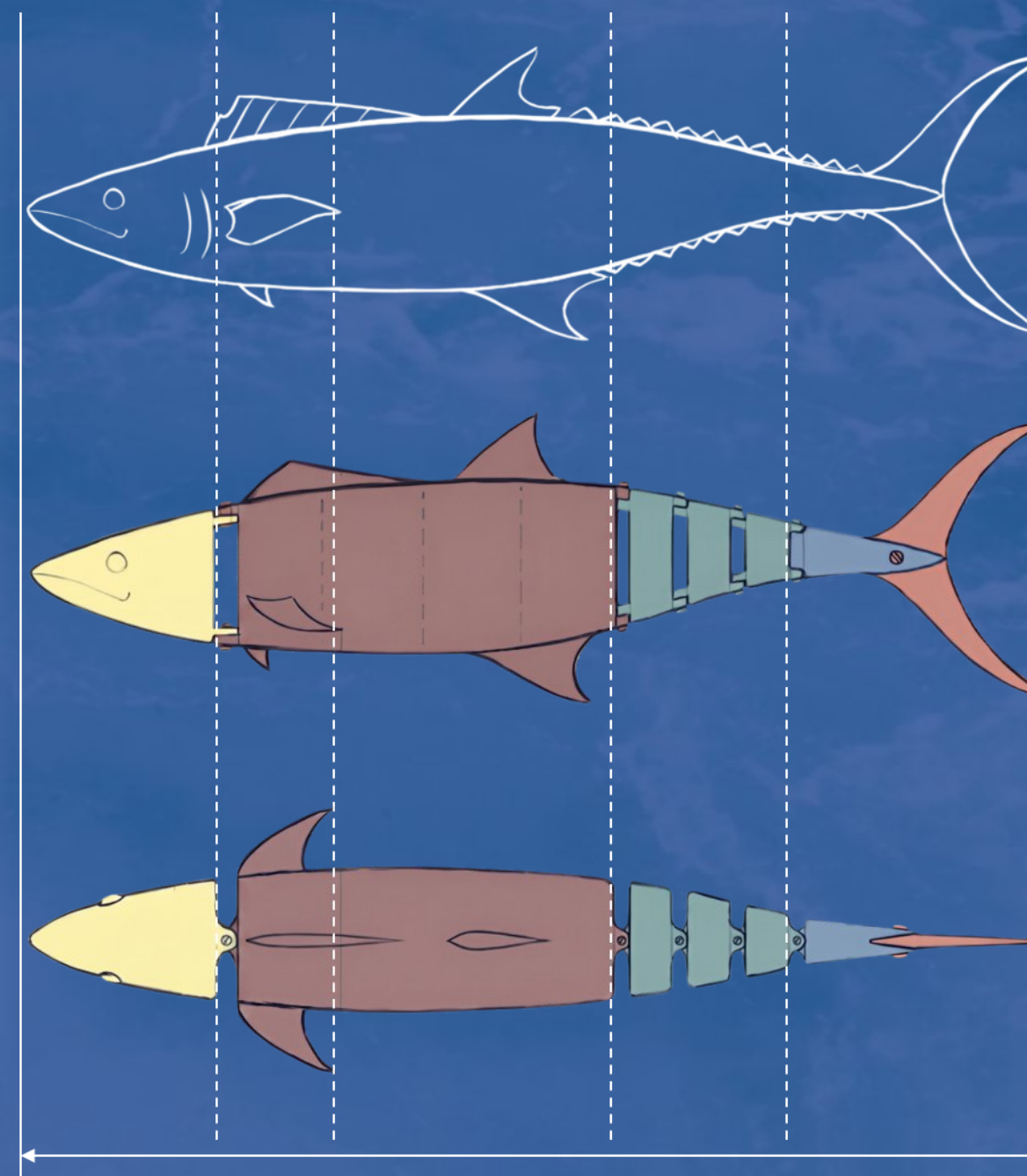
OpenFish [1]

Morphology Simplifications

- Simplify complexity but retain natural geometry, prioritizing straight-line movement
- Replace muscle-tendon networks with simplified mechanical alternatives

Structure Segmentation

- Active Head**
Rigid section at anterior tip driven by an independent servo motor to handle active steering and heading control
- Static Body**
Rigid central compartment containing and protecting the primary electronics, Arduino, and battery
- Passive Tail**
Flexible, segmented multi-link jointed portion of the tail propagating the undulatory traveling wave
- Active Tail**
Final tail linkage connected directly to the main motor drive mechanism to receive active mechanical power
- Rigid Caudal Fin**
Solid lunate-shaped profile attached to the active tail segment to for force transmission and forward propulsion



o2 Actuation Inspiration

The design implements a **linkage mechanism** to convert a motor's rotation into a propulsive traveling wave


Rotation to Undulation

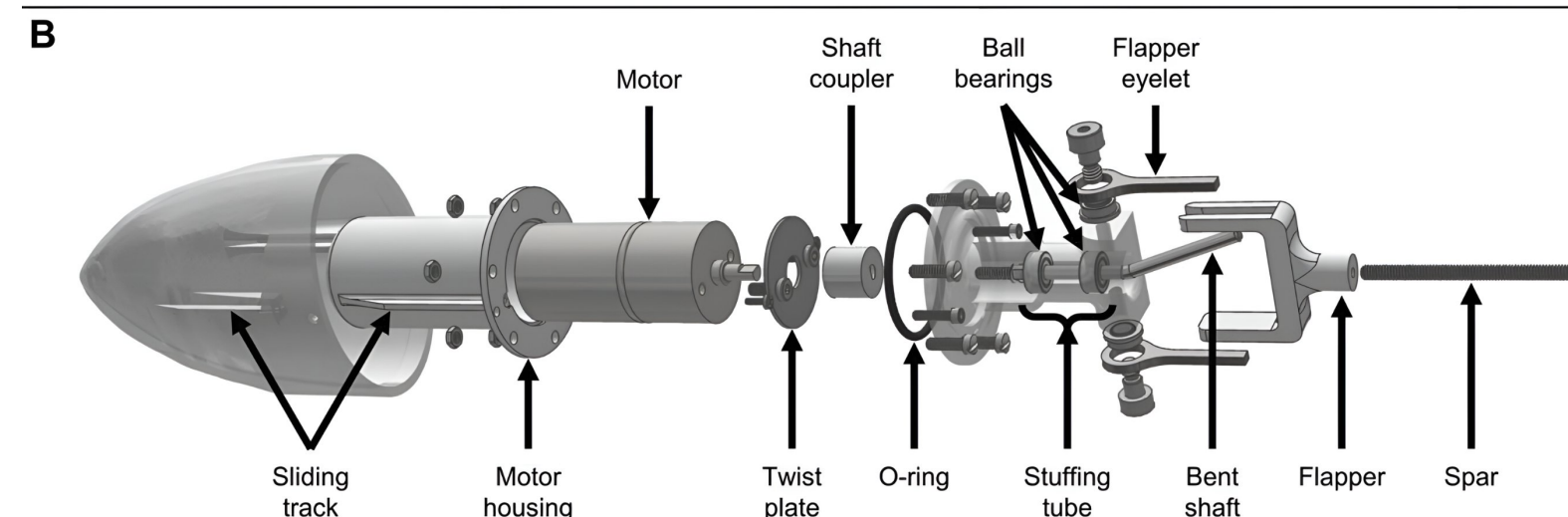
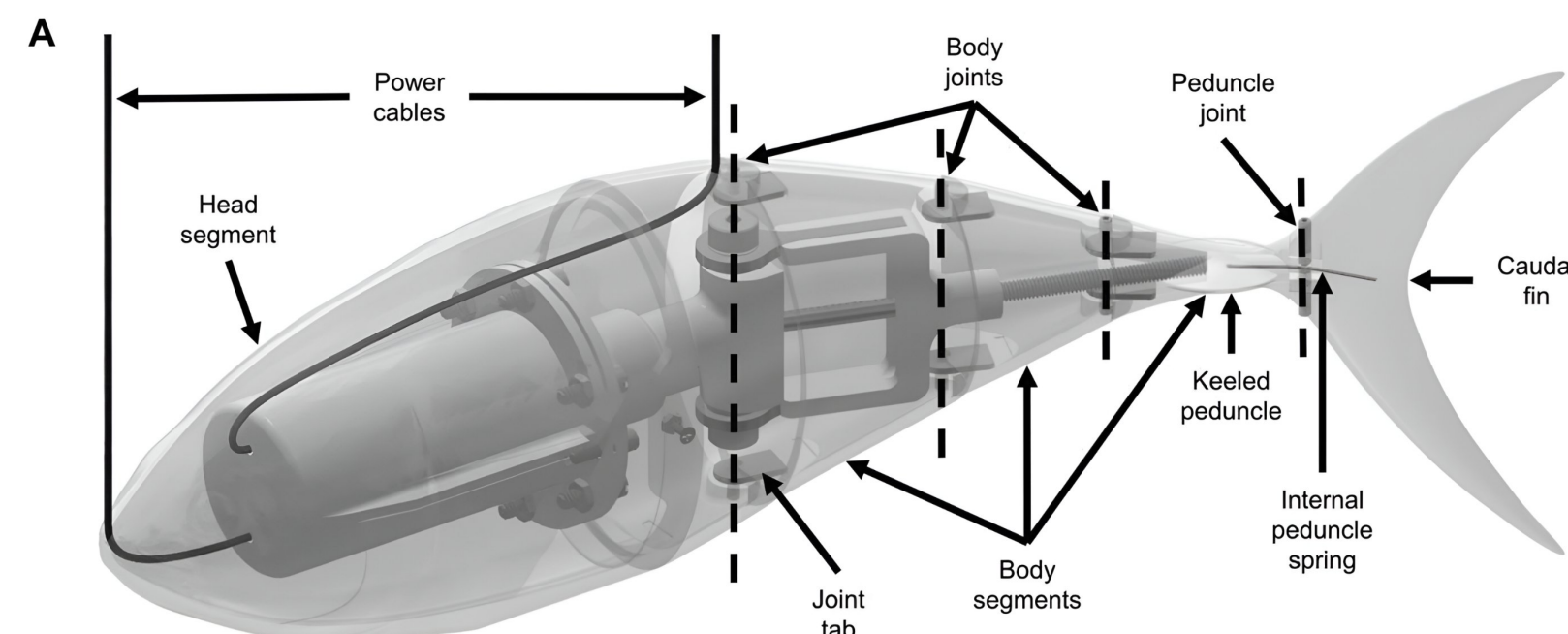
- A **drive mechanism** is used to translate a single motor's rotation into a wave
- The **synchronized** bent shaft, flapper, and spar mechanism **mimics** rhythmic, side-to-side tail beats


Force Transmission

- The drive mechanism consists of an **internal rigid spar**, which channels actuation force directly to the caudal fin to generate thrust
- **Passive tail segments follow smoothly**


Segmented Geometry

- Replaced complex muscles with a single-axis, rigid, segmented linkage mechanism to convert the single input into **lateral body movements**
- Approximated **undulatory wave profile**



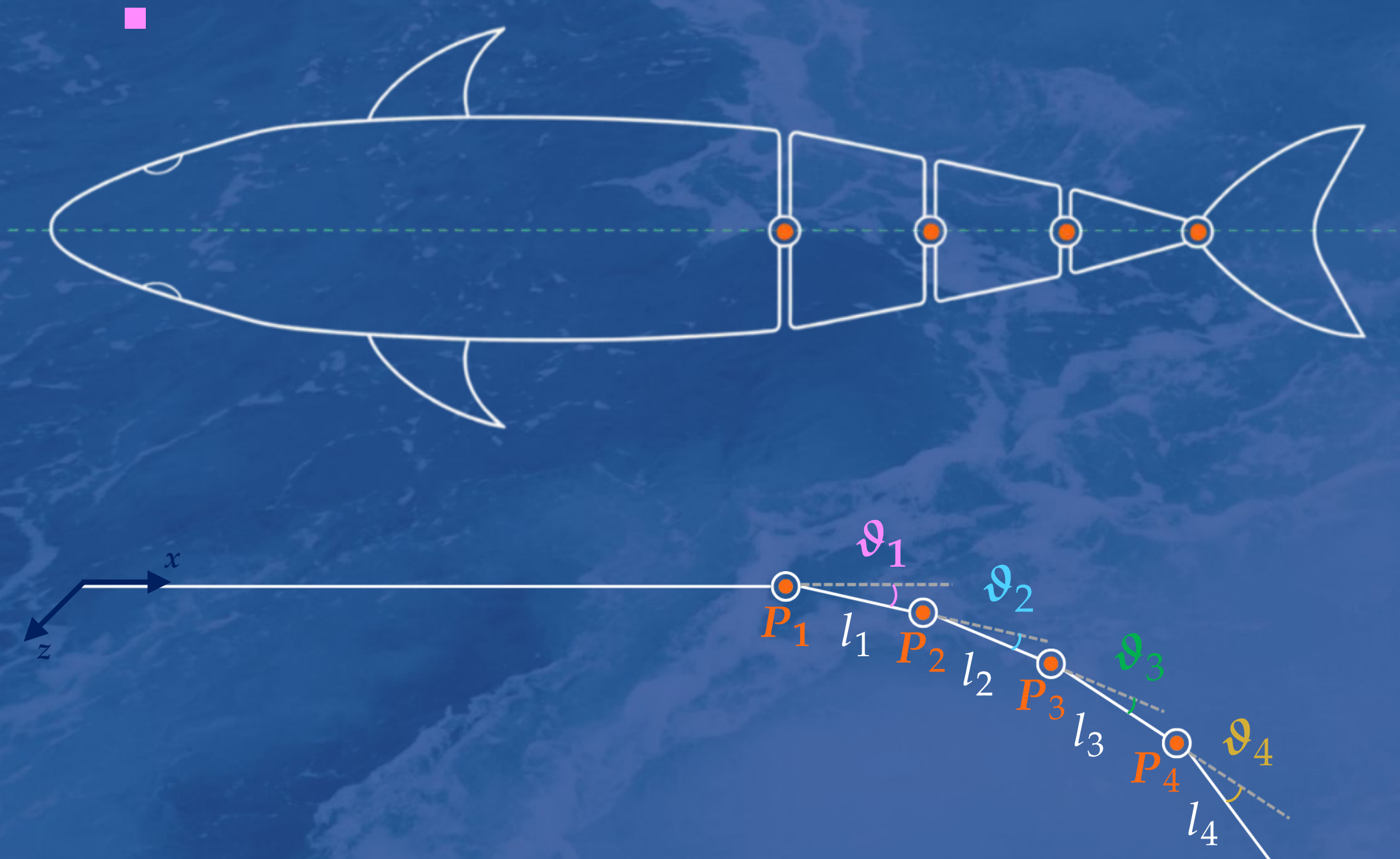
TunaBot Flex [2]

03.

kinematics & dynamics

o3 Kinematic model

Kinematic Diagram



- Rigid, non-actuated head and body
- Four-link actuated tail
- Relative joint angles: ϑ_1 , ϑ_2 , ϑ_3 , ϑ_4

o3 Motion law & Strouhal

Prescribed joint motion

$$\vartheta_i(t) = A_i \sin(\omega t + \varphi_i)$$

$$\Delta\varphi = -30^\circ \rightarrow \varphi = [0^\circ, -30^\circ, -60^\circ, -90^\circ]$$

$$\text{Relative amplitudes: } \alpha = [0.56, 0.67, 0.78, 1.00]$$

Strouhal amplitude scaling

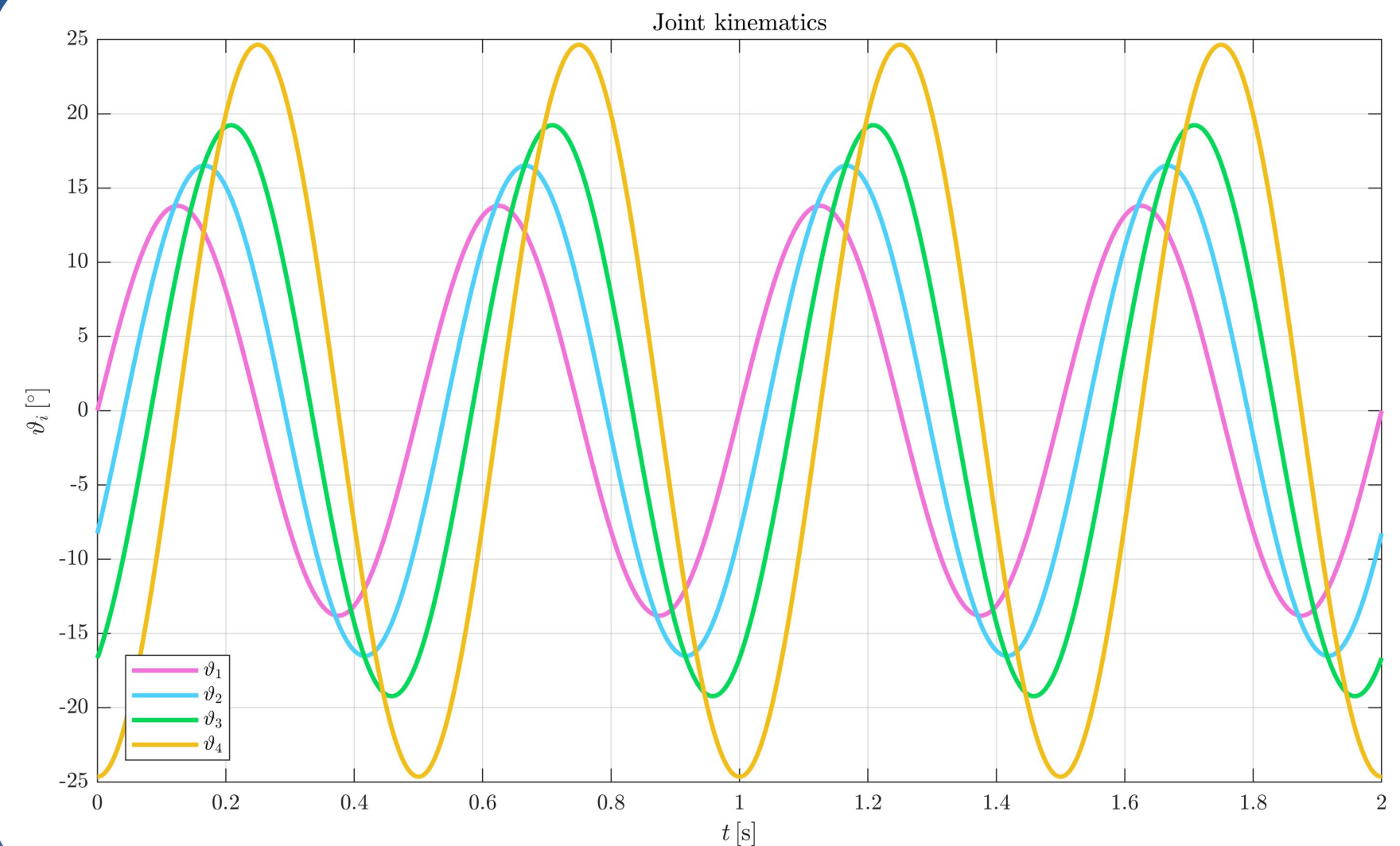
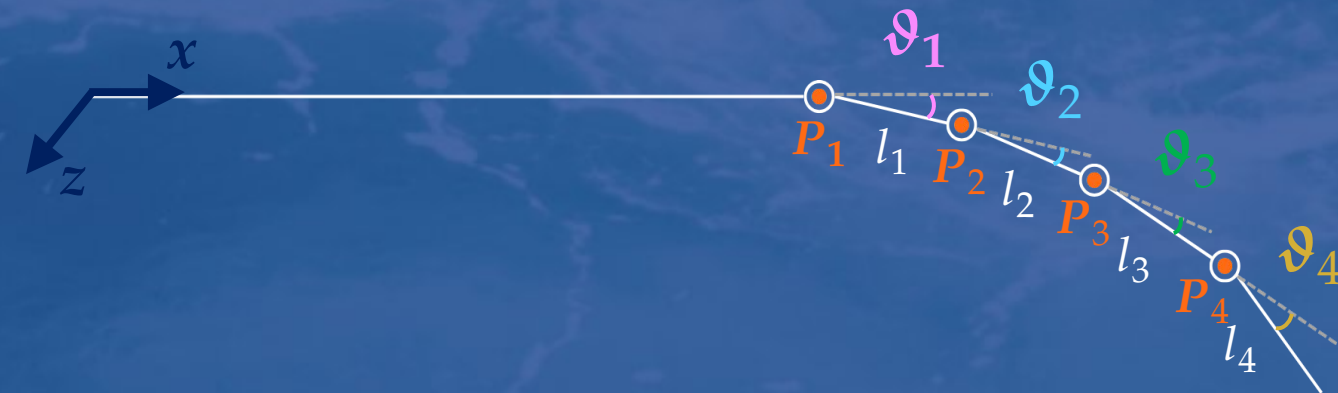
Chosen operating condition:

$$U = 0.70 \text{ m/s} \quad f = 2 \text{ Hz} \quad St_{\text{target}} = 0.40$$

$$St = \frac{2f A_{\text{tip}}}{U} \rightarrow A_{\text{tip, target}} = \frac{St_{\text{target}} U}{2f} = 70 \text{ mm}$$

Final matched joint amplitudes:

$$A_i = [7.81^\circ, 9.35^\circ, 10.88^\circ, 13.95^\circ]$$



o3 Centerline reconstruction

1. Relative joint angles

$$\vartheta_1(t), \vartheta_2(t), \vartheta_3(t), \vartheta_4(t)$$

2. Absolute link orientations

$$\theta_i(t)$$

$$= \vartheta_1(t) + \vartheta_2(t) + \dots + \vartheta_i(t)$$

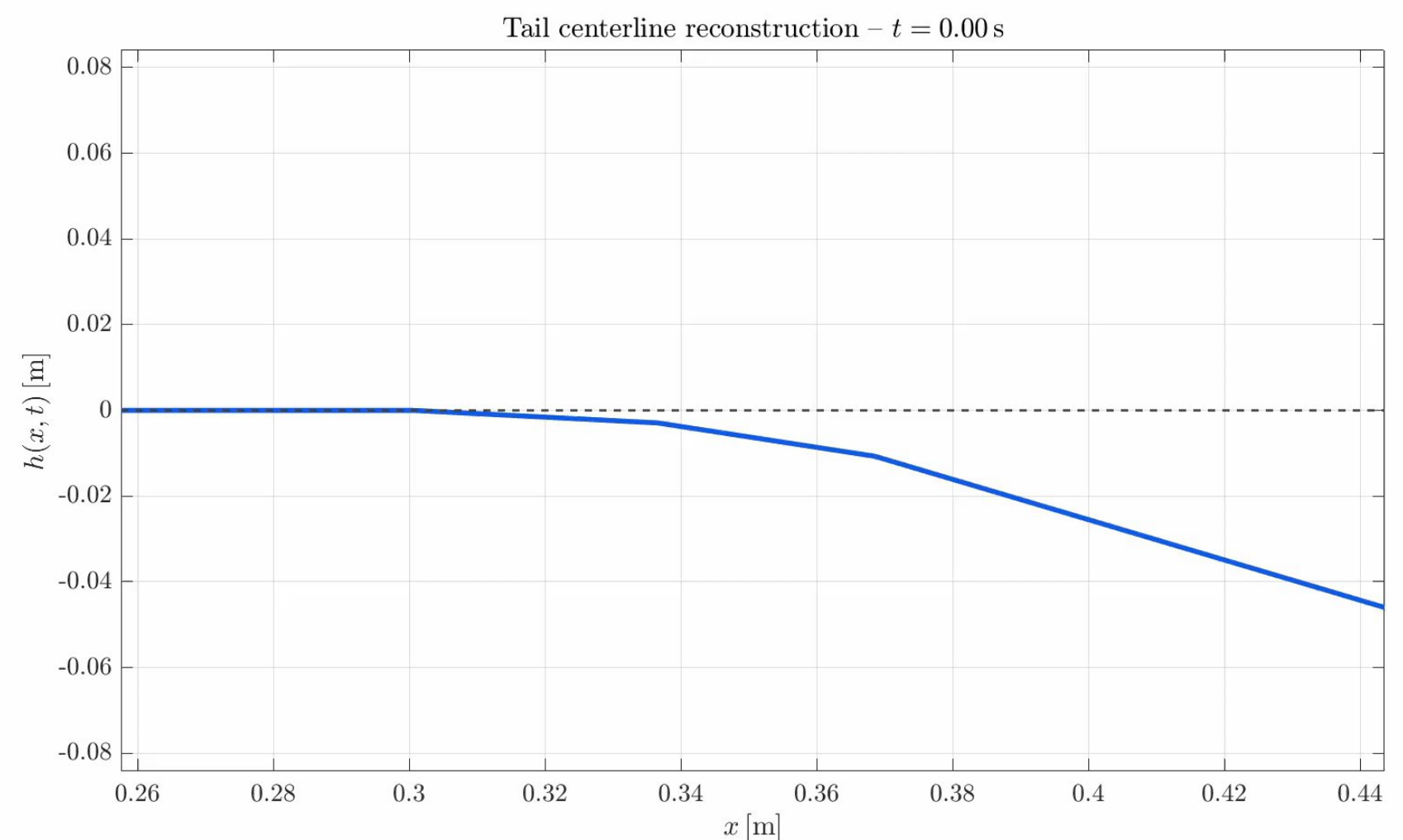
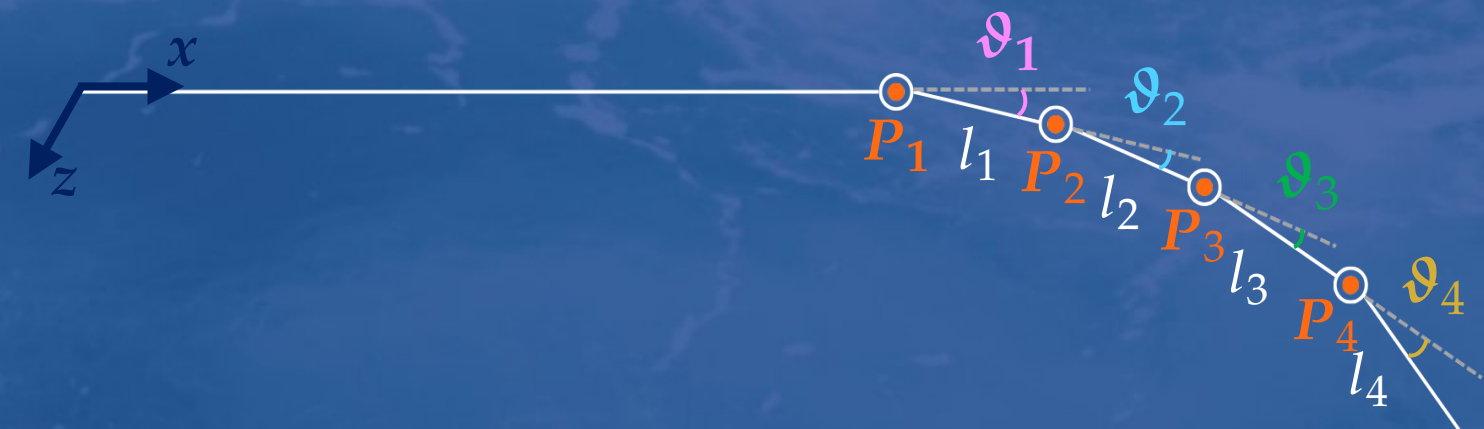
3. Tail-node positions

$$x_i(t) = x_{i-1}(t) + l_i \cos(\theta_i(t))$$

$$z_i(t) = z_{i-1}(t) + l_i \sin(\theta_i(t))$$

4. Continuous tail centerline

$$\{x_i(t), z_i(t)\} \\ \rightarrow h(x, t)$$



o3 Tail hydrodynamics

1

Inputs

- $h(x,t)$: tail centerline
- $A(x)$: equivalent area

2

LAEBT Model

- Inviscid flow
- Slender-body approximation
- Quasi-steady contribution neglected
- Added-mass-dominated force

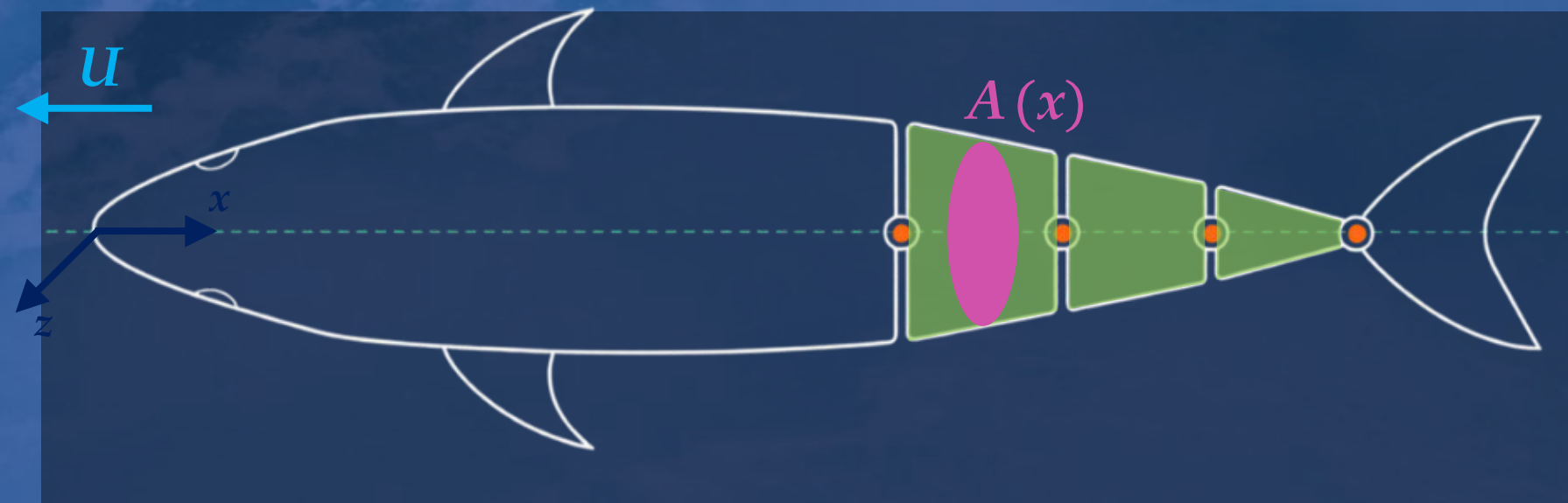
$$v_z = \frac{\partial h}{\partial t} + U \frac{\partial h}{\partial x}$$

Applied on the active tail only

3

Outputs

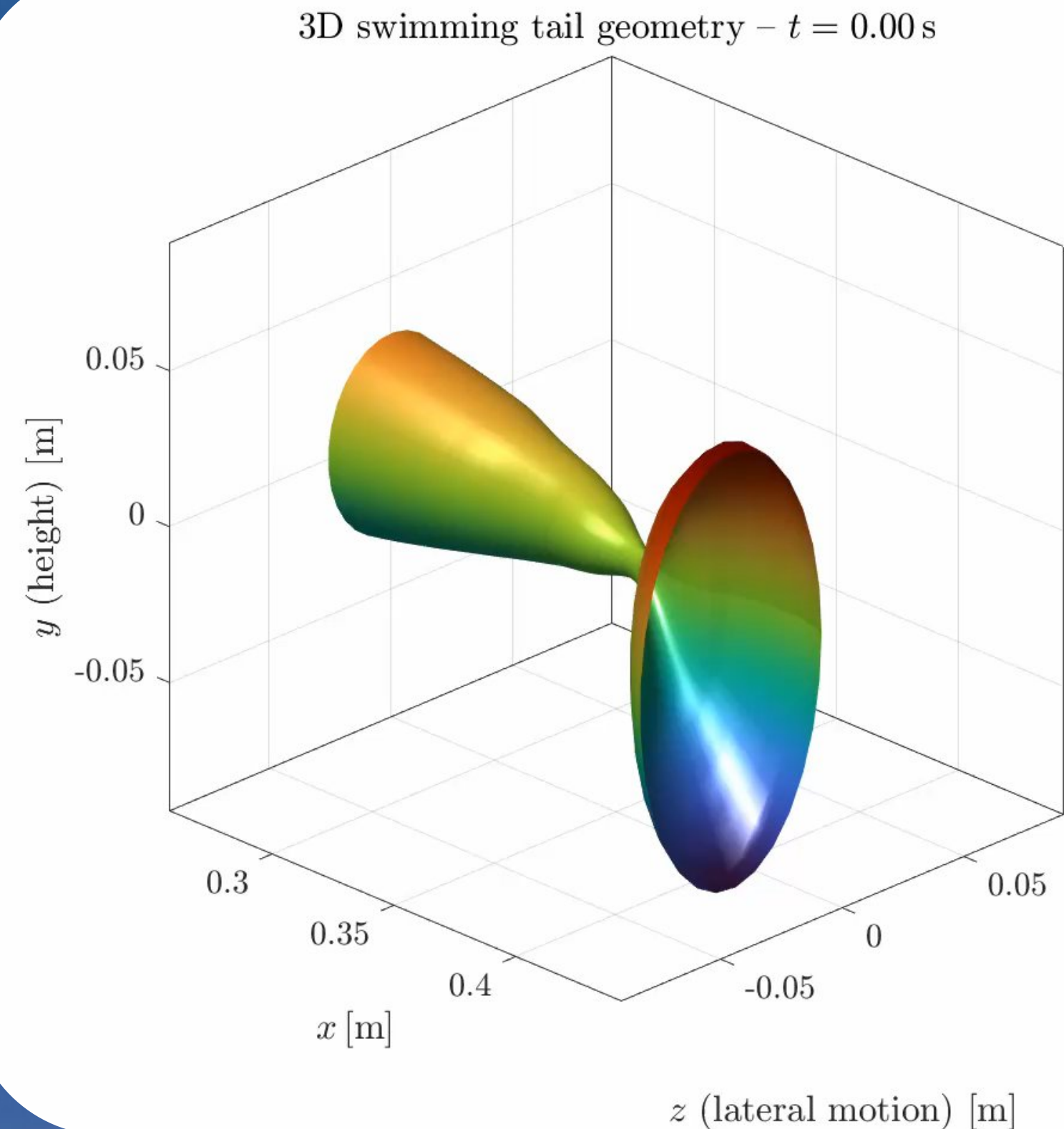
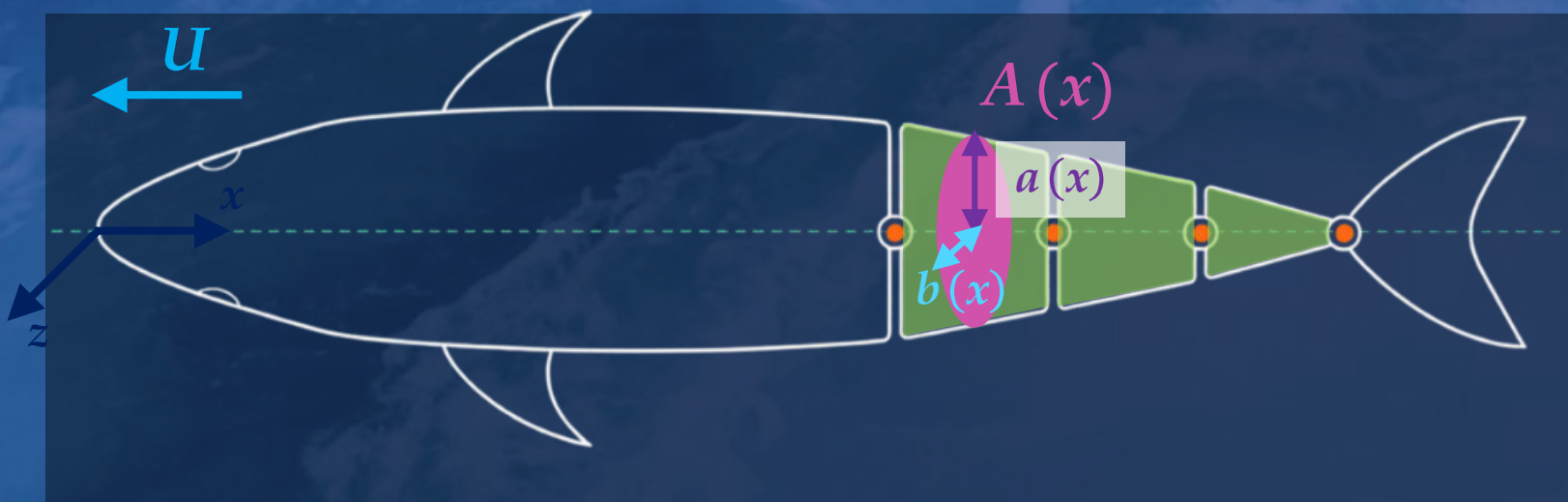
- $T(t)$: tail-generated thrust
- $P(t)$: hydrodynamic power



o3 Equivalent tail geometry

- Tail sections are approximated as **equivalent ellipses**
- The local equivalent area is

$$A(x) = \pi a(x) b(x)$$
- $A(x)$ is used together with the centerline motion $h(x, t)$ in the hydrodynamic model



o3 Thrust, power & efficiency

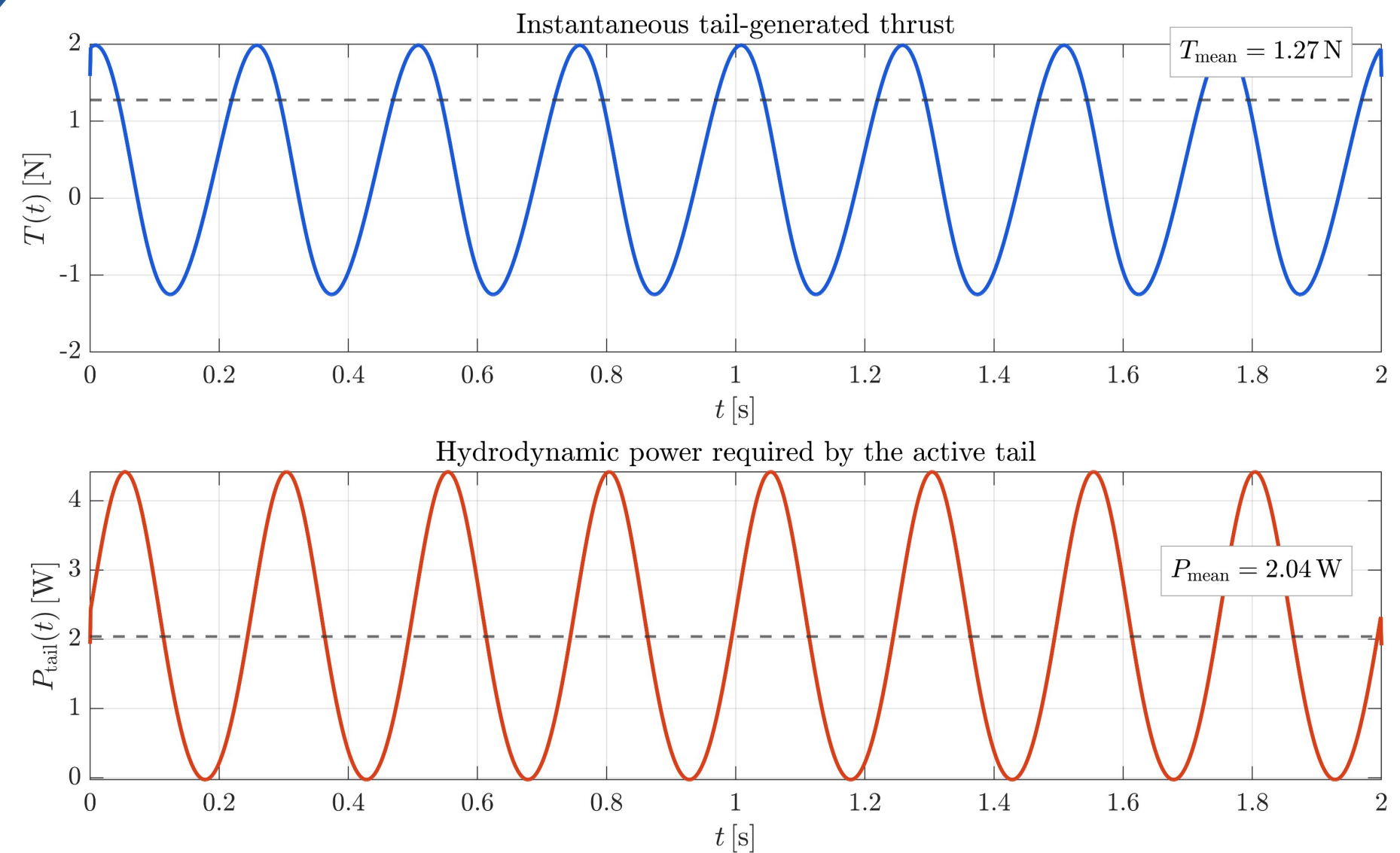
Cycle-averaged performance

$$T_{mean} = 1.27 \text{ N} \quad \text{mean tail-generated thrust}$$

$$P_{mean} = 2.04 \text{ W} \quad \text{mean hydrodynamic power required by the tail}$$

$$\eta = \frac{T_{mean} \cdot U}{P_{mean}} \rightarrow \eta = 0.437 \quad \text{propulsive efficiency}$$

$$P_{peak} = 4.42 \text{ W} \quad \text{peak hydrodynamic power (used as input for motor sizing)}$$



04.

Robotic design

o4 Simplifying Fish

We simplified the fish's morphology as shown in the Kinematics section to a structure made of a series of links.

We have the head link, body link, tail link and the caudal fin.

Morphology

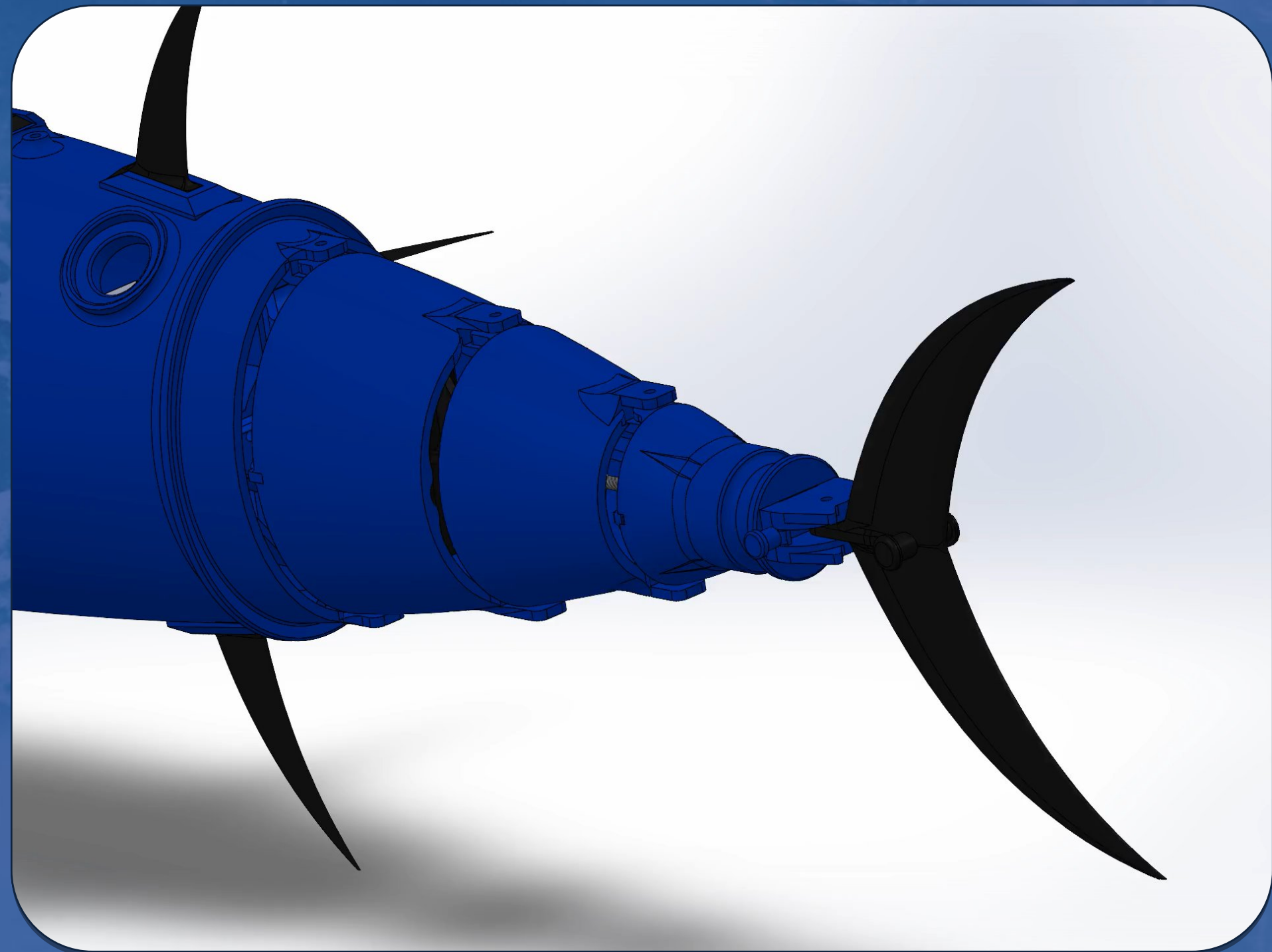


o4 The Core Mechanism

We use a version of the scotch-yoke mechanism. We have a single motor actuation with a bent shaft fitted in a slot providing oscillatory motion.



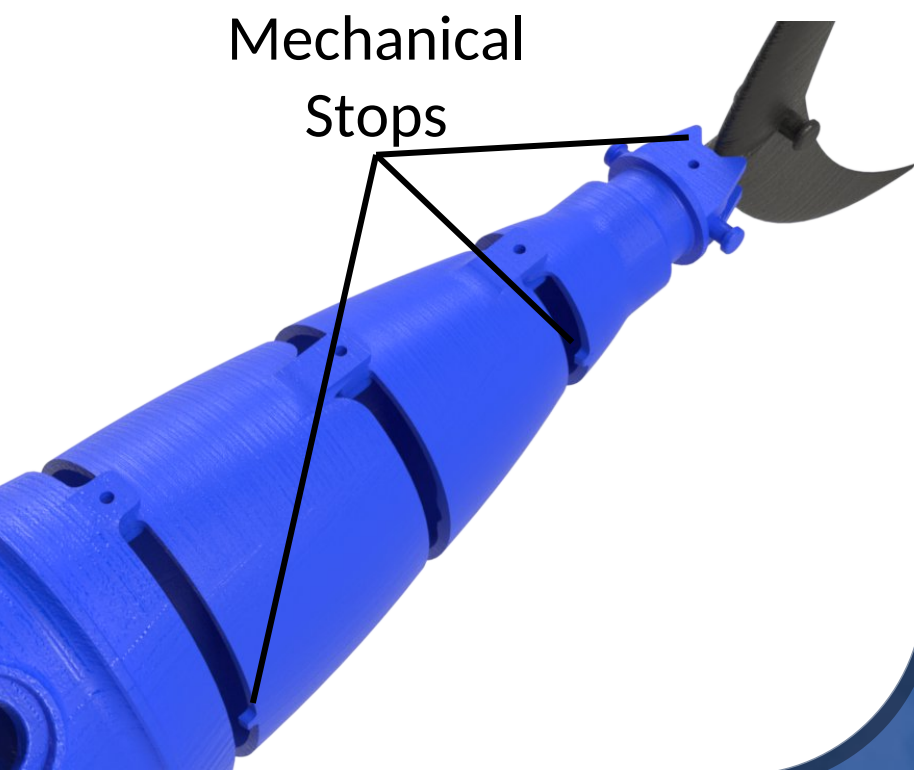
- High Tailbeat Frequency.
- Harmonic Tail Oscillation.
- Single Actuator.
- Adjustable Amplitude.



o4 Controlling the Amplitude

- Mechanical stops to control the angle of each link.
- Angle of bent shaft adjustable to control amplitude.

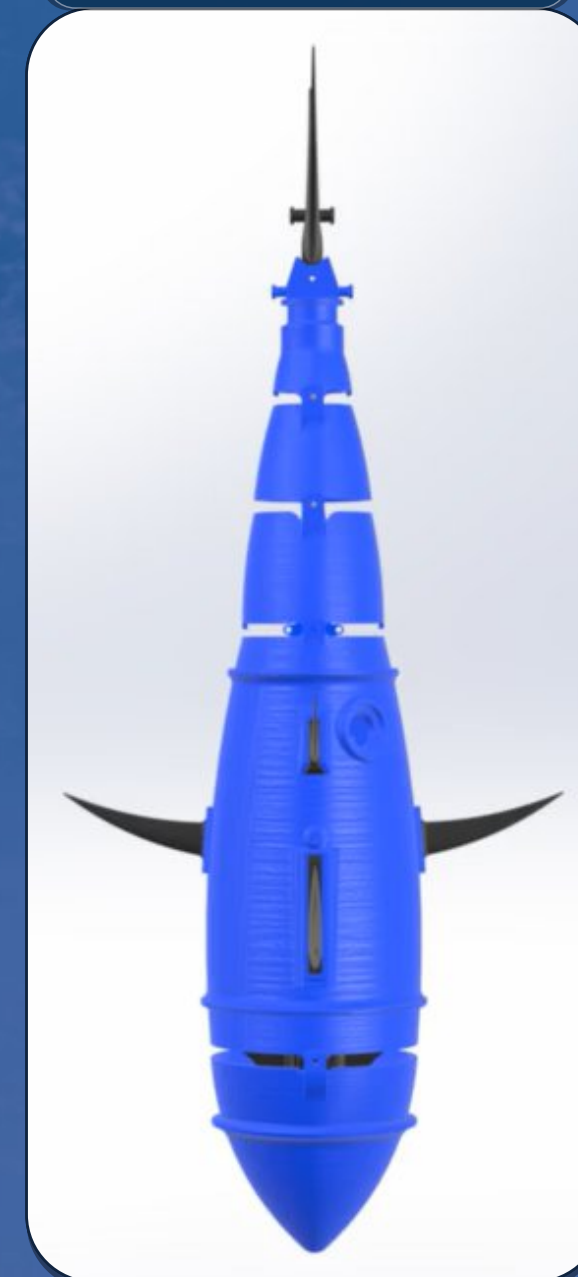
Mechanical Stops



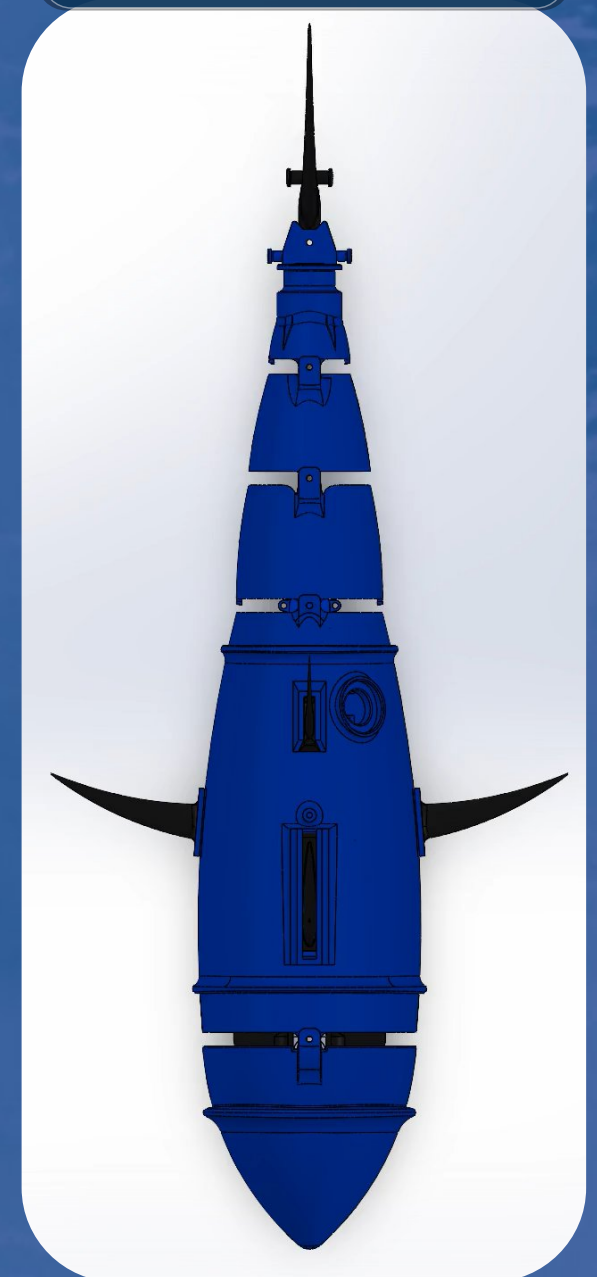
Total: 112 ° Sweep



Shaft Angle: 20 °



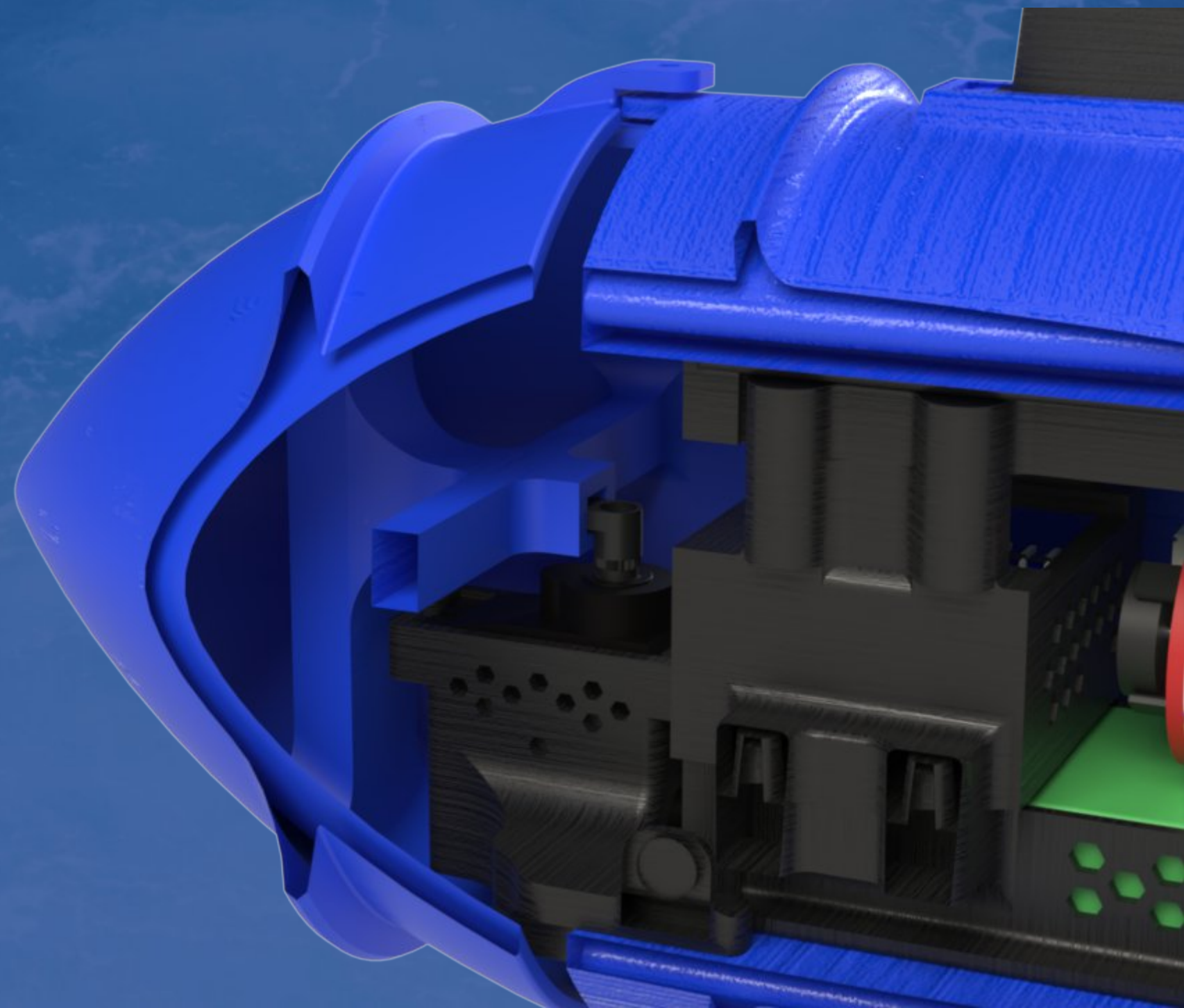
Shaft Angle: 10 °



o4 The Head Mechanism

We actuate the head to turn from side to side, and to ensure swimming in a straight-line if needed.

To do this we have a servo motor fixed in the body and connected via a flat horn to rotate the head.



o4Fin Design

In line with scientific references, such as TunaBot [2], the fins were designed following an **airfoil profile** [3].

NACA 0016



Caudal Fin



Primary driver of **forward propulsion**, utilizing high-frequency heaving and pitching motions to generate the main thrust force

Dorsal Fin



Passive **stabilizing surface** that manages pitch, counteracts roll, and supports straight-line lateral orientation

Anal Fin



Works with the dorsal fin on the hull bottom to **counteract roll** and ensure lateral stability

Lateral Fin (x2)



Passive stabilizer to **prevent** the body from **rolling** and maintain trajectory

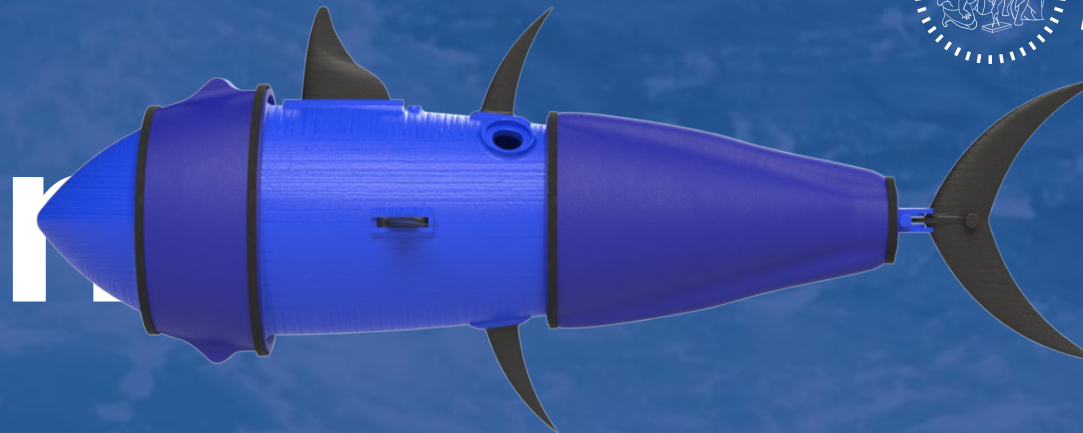
Solid, rigid fins maximize propulsive thrust transmission directly into the water by preventing energy losses caused by bending and damping

o4 Component Sizing and

Selection

Component	Selection Criteria
DC Motor	High torque-to-weight ratio, Integrated encoder, Optimal gear reduction, Compact
Battery	Appropriate operating voltage, High energy density (Ah/g), Compact form factor
Servo Motor	Adequate stall torque, Fast actuation speed, Precise angular resolution
Electronics	Processing speed, Current rating, Sensor accuracy, I/O pins

o4Material Selection



Material	Components	Reason
PLA	External body, Fins	Lightweight, easy to print, good stiffness
ABS	Motor Supports	Higher strength and impact resistance
Low-Carbon Steel	Bent Shaft	Ductile, high fatigue resistance
Carbon Fiber	Spar	High stiffness-to-weight ratio
Brass	Motor Coupling	Corrosion resistant, easily machined
Ecoflex 00-10 Silicone	Tail and Neck Sleeve	Waterproof, high elasticity seal
Elastic Latex Membrane	Tail and Neck waterproofing	Extra waterproofing

o4 Manufacturing

Our manufacturing process consisted of:

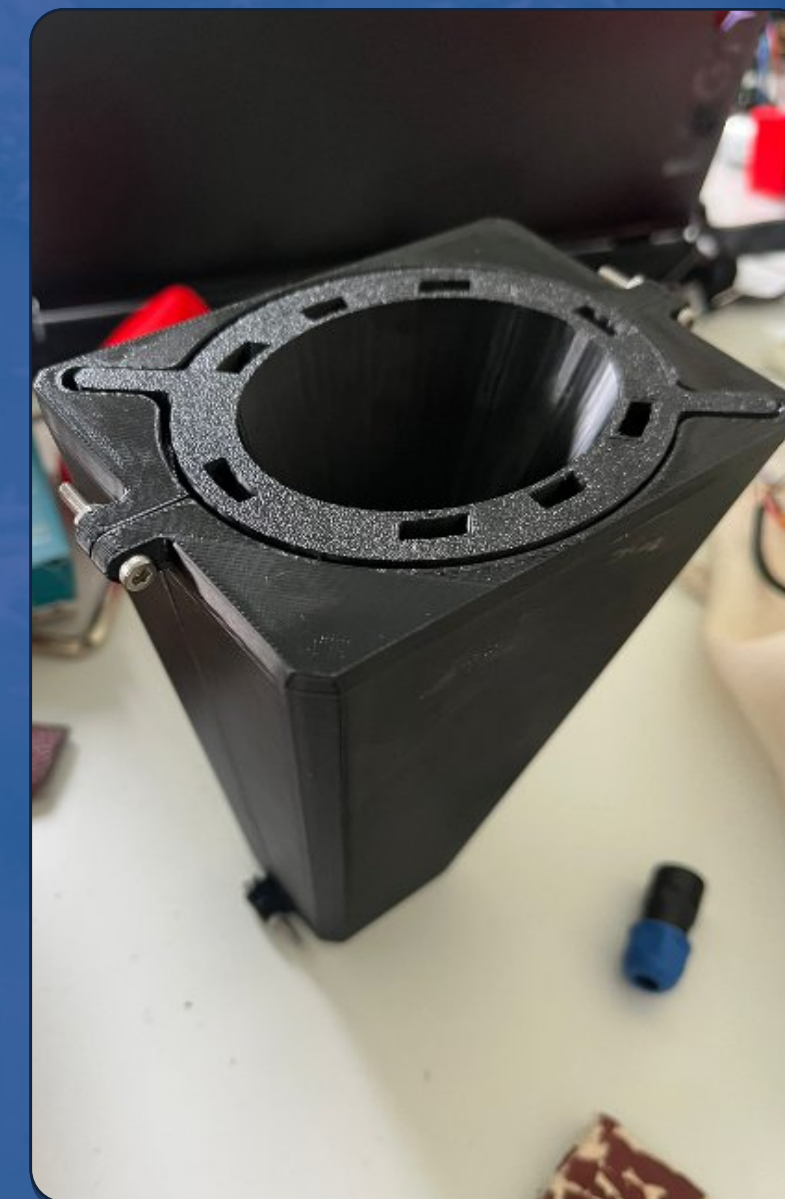
- 3D printed components
- 3D printed mold for silicone sleeve fabrication.
- Eco-Flex mold
- Manual Bending of steel shaft to achieve desired angle



3D Printed Tail
Segments

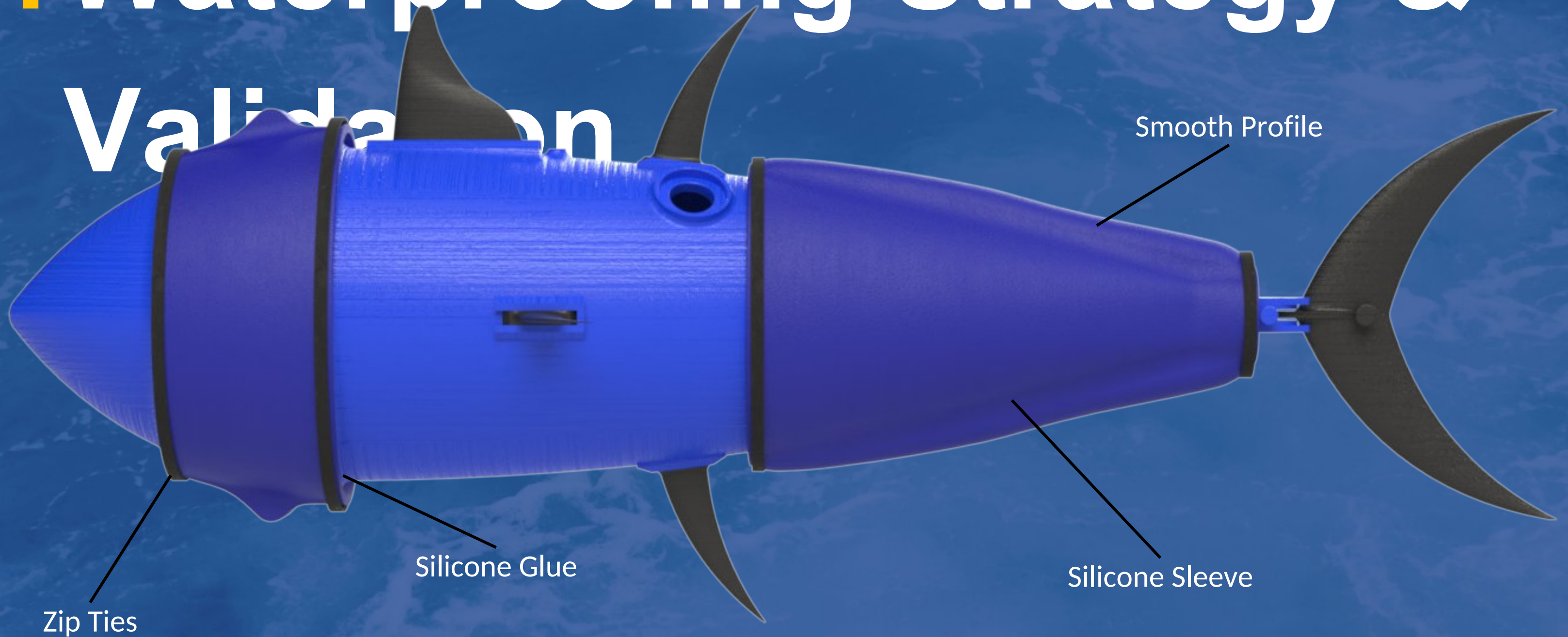


CAD Mold Design



3D Print Mold

o4 Waterproofing Strategy & Validation



Validation

Single Part Testing



Assembly Testing



Inspection



Iterative redesign

o4 Buoyancy

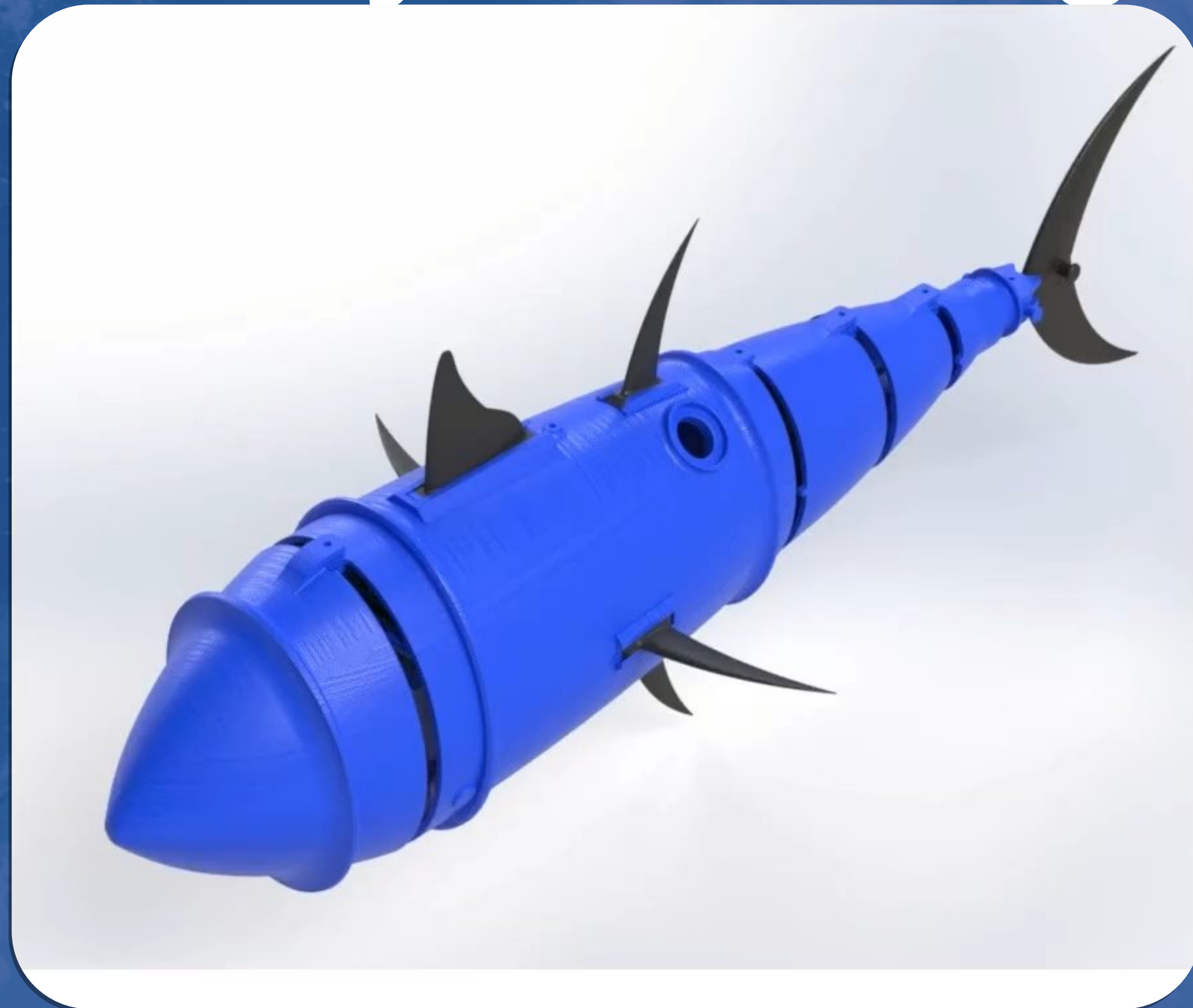
To achieve a neutrally-buoyant fish we incorporate:

- Enclosed design providing internal air volume
- Weight of the components.
- Additional steel weights [5 g per ballast]



Fish Weight	Added Weight	Total Weight	Total Internal Volume
750 g	780 g	1530 g	901.7 cm ³

o4 Assembly and integration



05.

electronics & control

o5 Electronic architecture

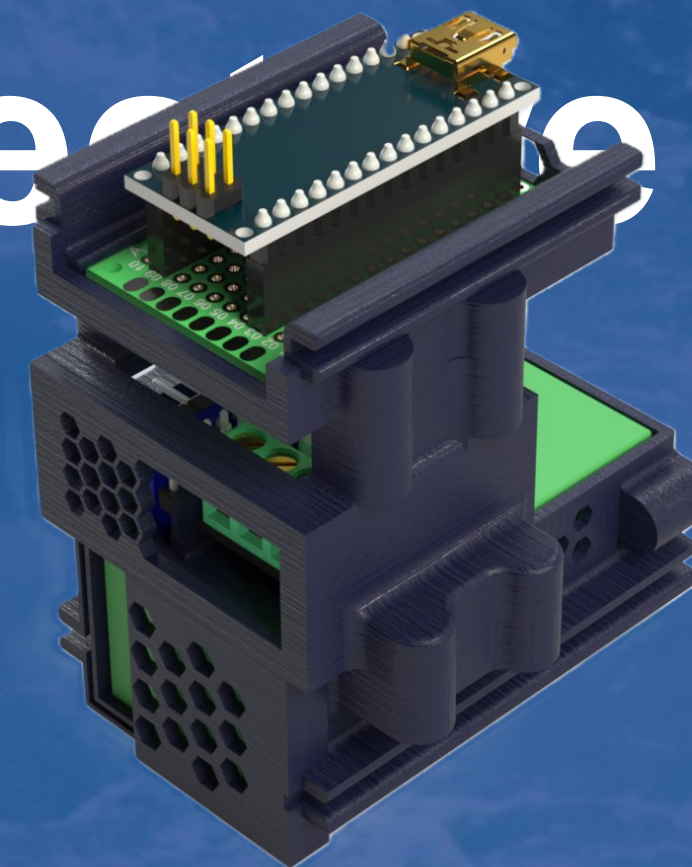
Connections and Components

Battery

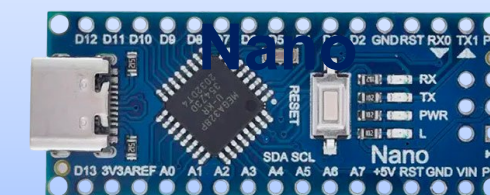
Arduino Nano

DC Motor
Encoder

Steering
Servo
IMU



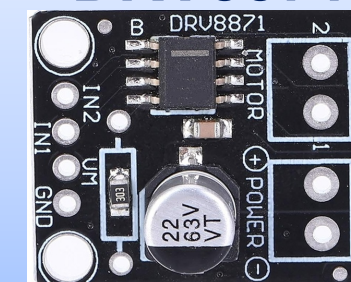
Arduino



Battery



DRV8871



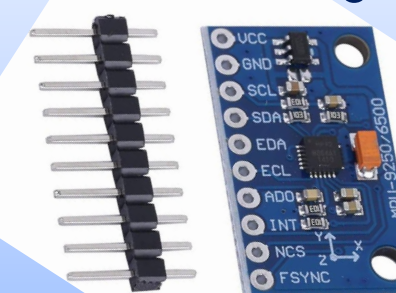
Button



DC motor & Encoder



250



Servo
motor



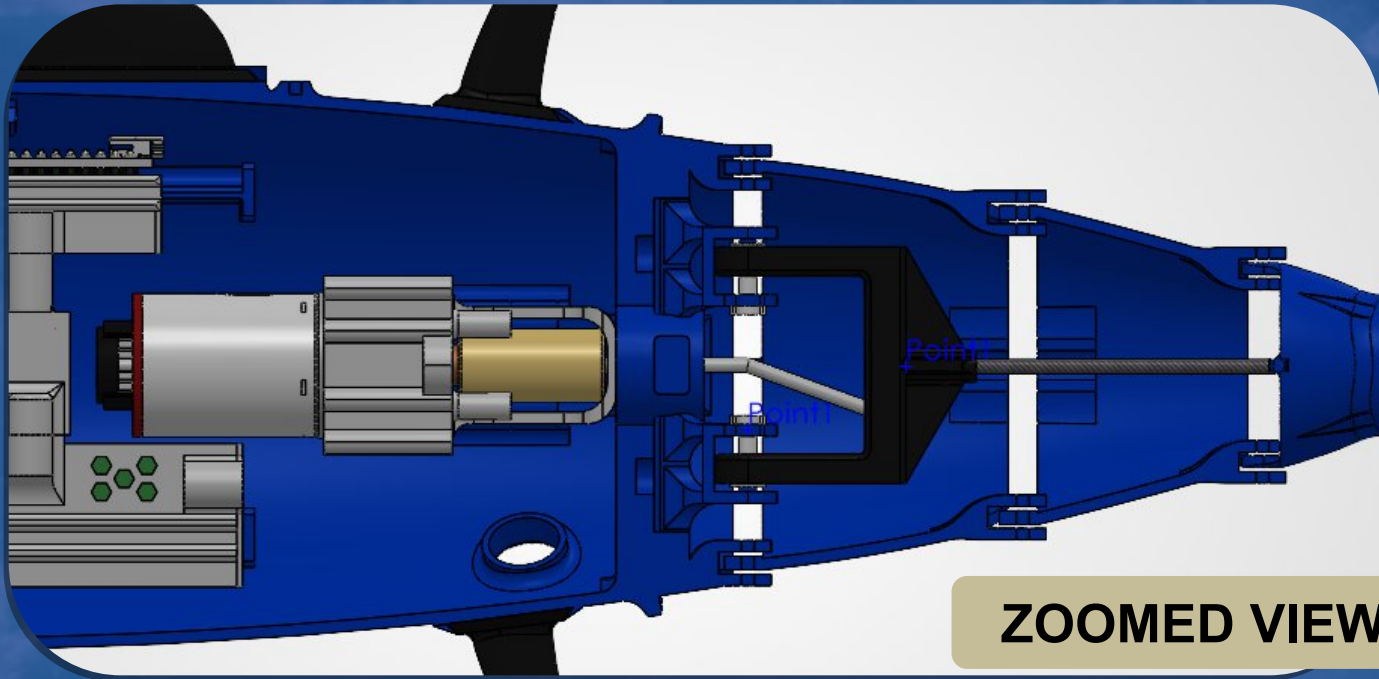
o5 Motor Selection &

Actuation system Integration

- **Motor:** ChiHai CHR-GM25-370ABHL Brushed DC Motor (12 V, 350 RPM)
- **Gearbox:** Integrated GM25 Reduction Gearbox
- **Encoder:** Integrated Incremental Encoder (CHR-GM25-370ABHL Series)

$$P_{mean, power} = \tau_{peak} * \omega = 0.51 * \frac{2 * \pi * 350}{60} = 18.7W$$

Motor Torque = 1.17 Nm



Motor Type	Brushed DC Geared Motor with Encoder
Nominal Voltage	12 V
Rated Speed	350 RPM
Mass	110 g
Output Current	60-200 mA
Diameter	25 mm
Integrated Encoder	Yes

Sizing from dynamics

$$P_{peak, hyd} = 4.42 \text{ W} \quad \text{peak hydrodynamic power}$$

$$\tau_{peak, hyd} = 0.35 \text{ Nm} \quad \text{peak hydrodynamic torque}$$

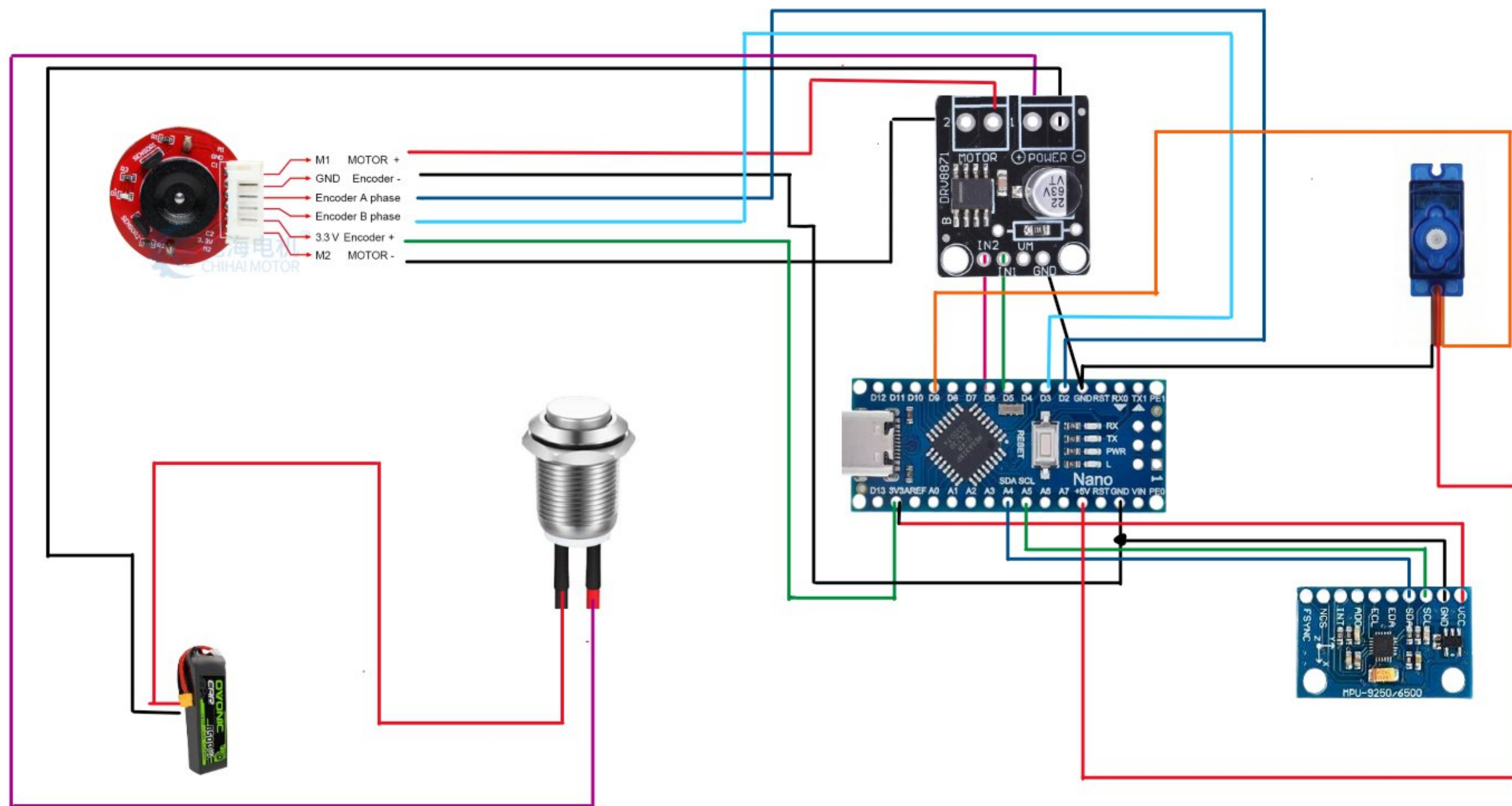
$$K_{loss} = 1.20 \quad \text{silicone losses}$$

$$SF = 1.20 \quad \text{safety margin}$$

$$P_{req} = SF \cdot K_{loss} \cdot P_{peak, hyd} = 6.36 \text{ W} \quad \text{required motor power}$$

$$\tau_{req} = SF \cdot K_{loss} \cdot \tau_{peak, hyd} = 0.51 \text{ N} \quad \text{required motor torque}$$

o5 System Circuit



o5 Software architecture

Modules

Mode Manager

- OFF/ Normal/ Fast modes
- Magnetic tap/ hold detection

Propulsion Controller

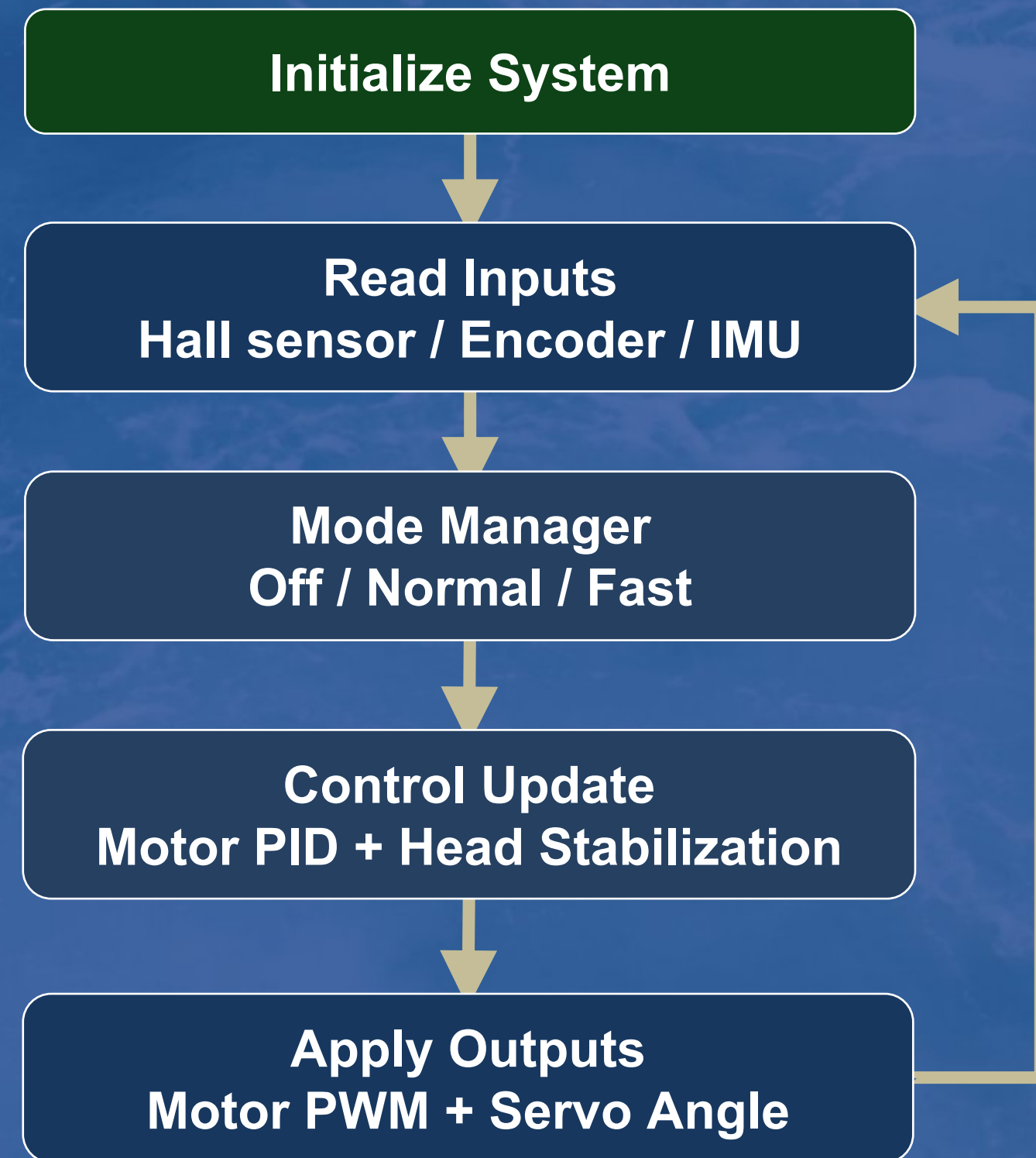
- Soft-start ramp
- PID motor speed control
- PWM motor command

Feedback Module

- Encoder speed acquisition
- IMU gyro acquisition

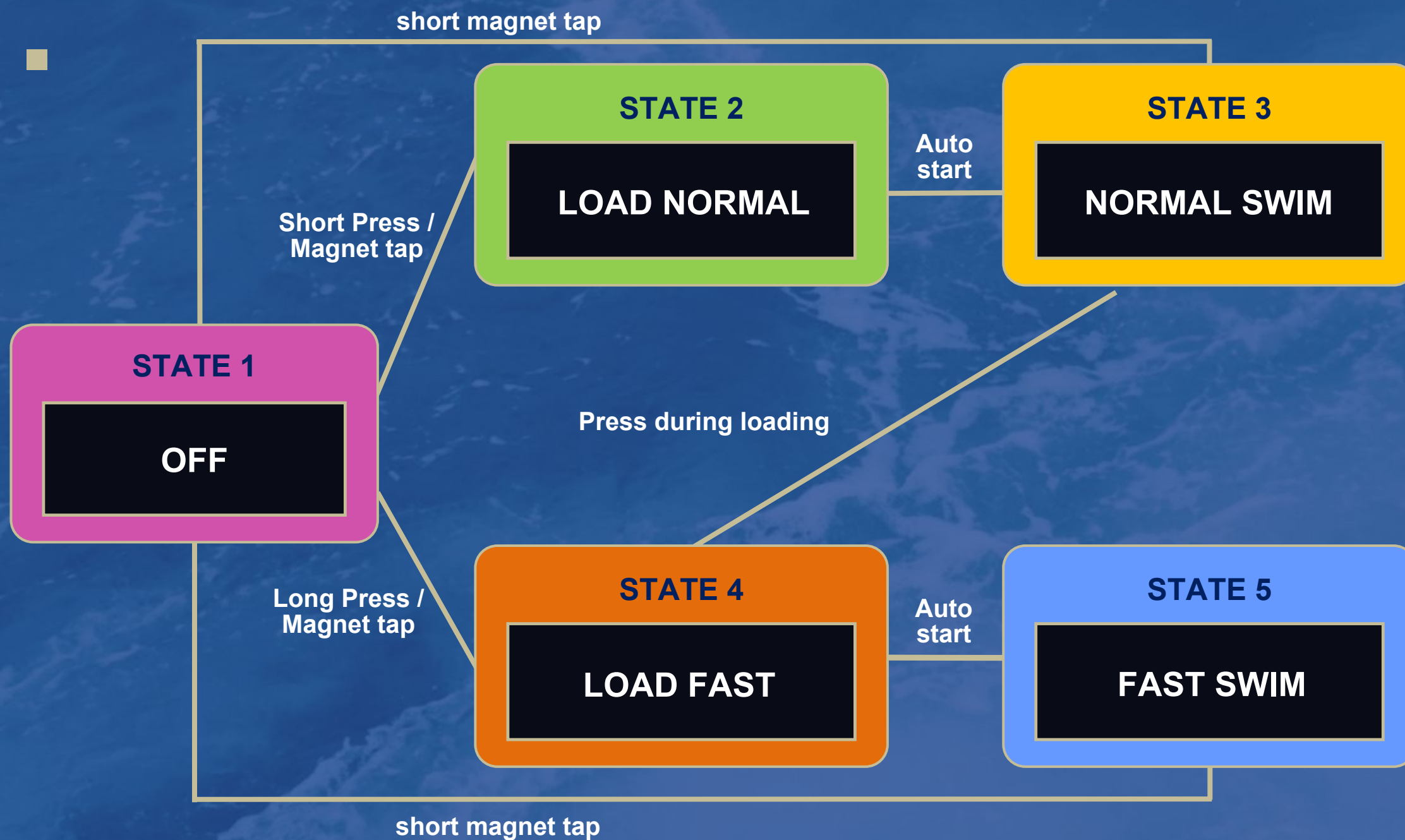
Steering Controller

- Drift angle selection
- Servo head correction



o5 Machine control

Finite-state machine for switching between normal and fast propulsion modes.



States

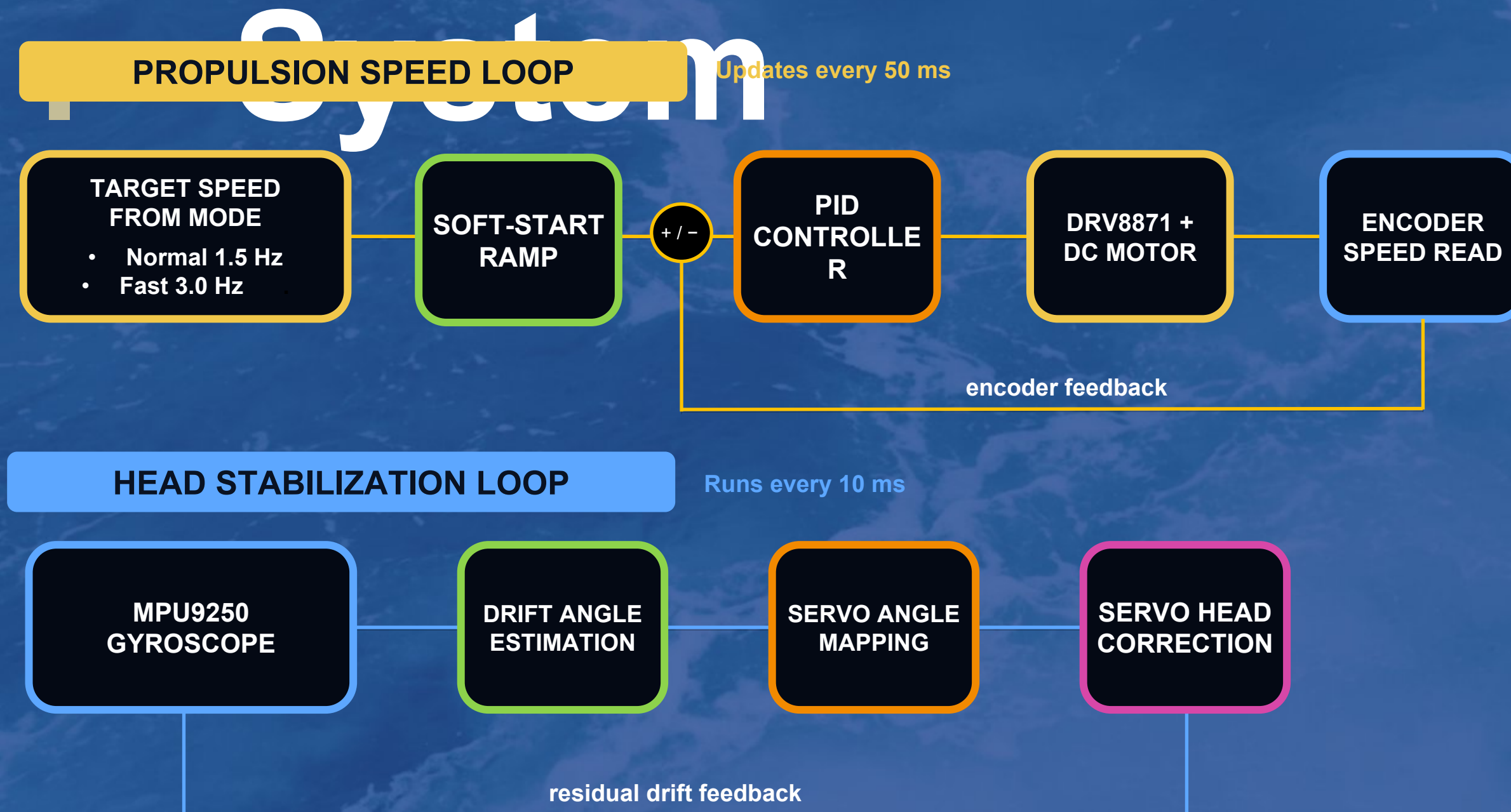
- OFF
- Loading Normal Mode
- Normal Swimming
- Loading Fast Mode
- Fast Swimming

Events

- Short magnetic tap
- Long magnetic hold ≥ 3 s
- Soft-start complete
- Short magnetic tap while swimming
- Magnetic tap during loading

o5 Closed-Loop Control

Arduino-based control of propulsion and head stabilization.



Control Summary

Control loops

- Propulsion speed loop
- Head stabilization loop

Sensors

- Encoder: motor speed
- MPU9250: body rotation

Actuators

- DC motor via PWM
- Servo head correction

The Arduino code keeps the selected swimming frequency stable while independently correcting body drift through the servo head.

06.

testing & RE\$ults

o6 Dry Motion VALIDATION

Tail actuation test outside water using the silicone body sleeve.



Test objective

Validate the mechanical transmission and tail oscillation before water exposure.

Verified functions

- Motor-driven tail actuation
- Linkage motion inside silicone sleeve
- Normal and fast mode response
- Servo/head movement

Result

Expected oscillatory tail motion was achieved in air.

o6 Waterproofing & Buoyancy

Immersion check used to verify sealing and floating behavior before propulsion testing.

Test

Observations during Immersion

- ✓ Sealed sleeve stayed intact
- ✓ Electronics isolated
- ✓ No visible leakage observed
- ✓ Stable floating posture
- ✓ Ready for propulsion test



Stable waterline

Sealed body

Float check

Initial Immersion and Flotation Check

o6 Initial In-Water Trial: Caudal

Low-speed test used to observe natural body response before adding stabilizing fins.



Test objective

- Observe body stability in water
- Check if tail oscillation produced forward motion
- Identify rolling/yawing behavior without side fins

Observation

- Slow forward motion was generated
- Directional stability was limited
- Missing fins affected balance and straight-line motion

Design insight

Side fins were needed to improve passive stability before final swimming tests.

o6 Thrust Readings

Comparison between theoretical and experimental thrust at **1.5 Hz**

Peak Hydrodynamic Thrust (Theoretical)

■ 1.2 N

Peak Dynamic Load (Experimental)

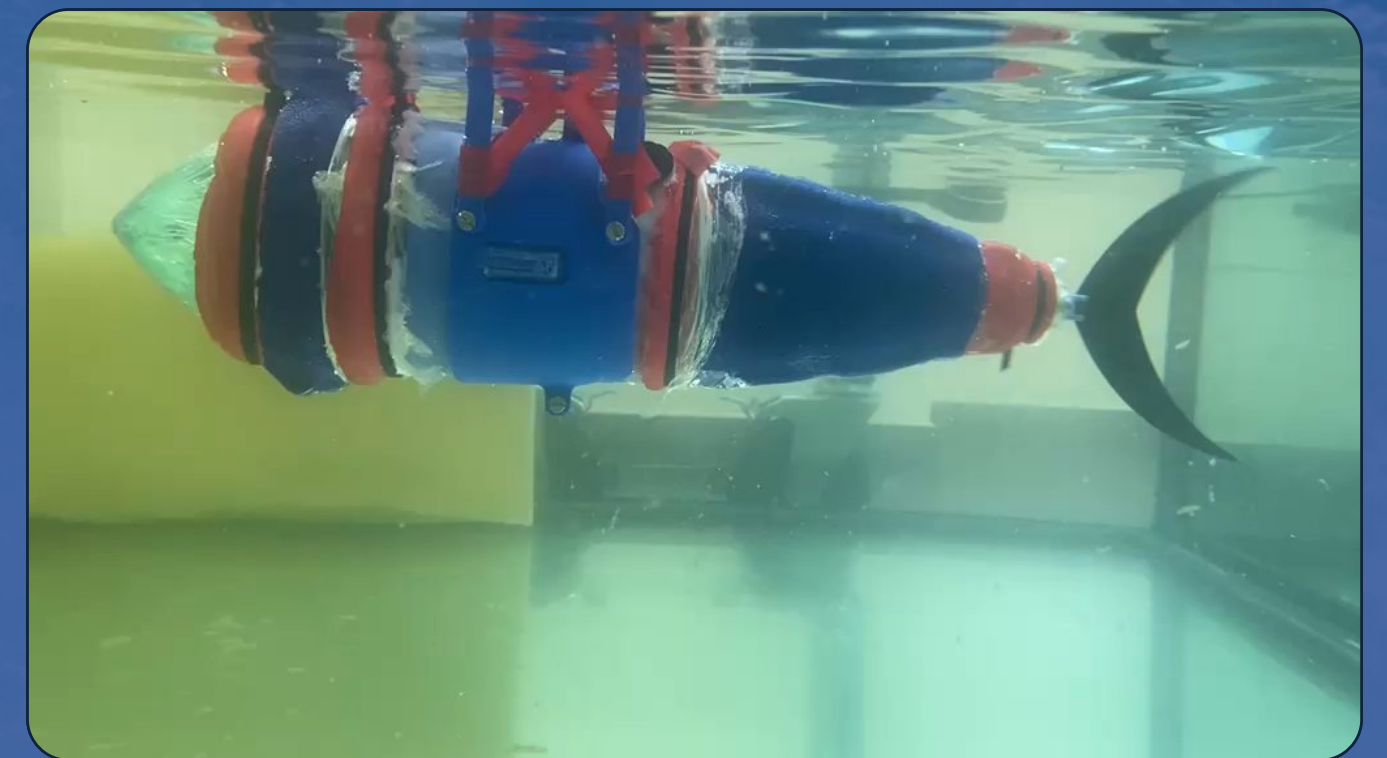
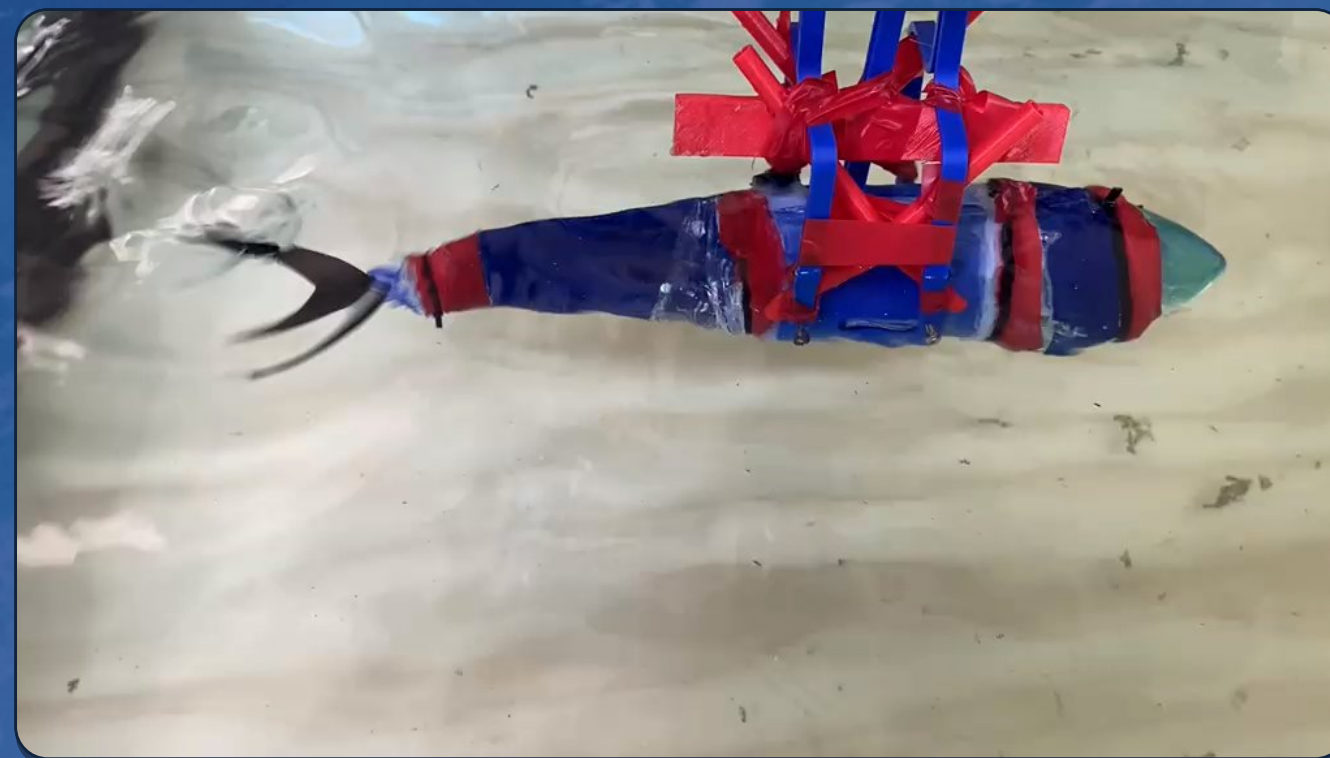
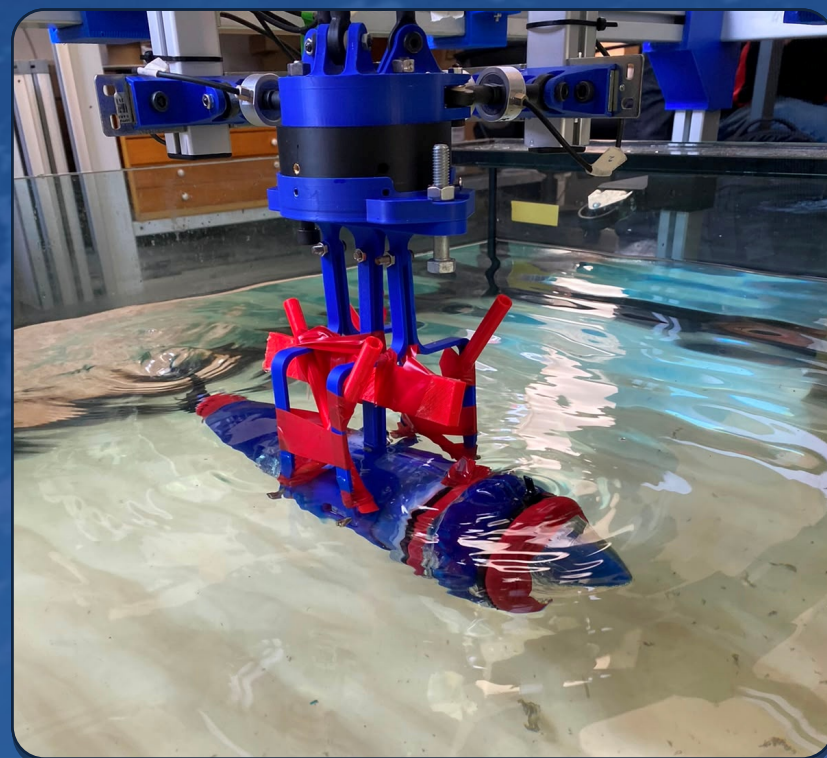
1.38 N

0.18 N
(15%)

Reasons for Difference

- **Theoretical:** Calculates pure fluid forces only
- **Experimental:** Sensor captures fluid + tail inertia.
- **+15%:** Heavy tail momentum drives peaks higher.

Experimental Setup for Thrust Measurement



o6 Final In-Water Swimming

The fish was deployed in a testing tank and successfully completed its **straight-line swimming goal**.

■ Test



At 1.5 Hz (30% Max Speed)

0.33 m/s

0.73 BL/s

Thank you for your attention

